MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

**Project** **Mansourah Massarah Gold Project**
**Customer:** **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

**Title:** **DESIGN CALCULATION**
**4000 PROCESS PLANT: ENGINEERING OFFICE CUM CCR**
**DESIGN OF BUILDING STRUCTURE**

**Document Number** **MD-216-4000-EG-CV-CAL-1080**
**Revision:** **D01**
**Revision Date:** **30.10.2023**
**Status:** **AS BUILT**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Outotec Document Number: | 900707 | GAU01 | AN03 | CED01 | 40005 | OU100683341 | 2 |
| L&T Document Number: | O19045 | 900 | C | K | 24040 | 001 | 2 |

**REVISION HISTORY**

| Rev | Date | Revision Text | Prepared By | Reviewed By | Approved By | Employer |
|---|---|---|---|---|---|---|
| A01 | 10.06.2020 | FIRST ISSUE | MF | AVG | SJM | |
| C01 | 06.07.2020 | ISSUED FOR CONSTRUCTION | MF | AVG | SJM | |
| D01 | 30.10.2023 | AS BUILT | MF | AVG | SJM | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

# TABLE OF CONTENTS


DEFINITIONS AND ABBREVIATIONS ..... 4
1. INTENT ..... 5
2. EXECUTIVE SUMMARY ..... 5
3. SCOPE ..... 5
4. DESIGN METHODOLOGY & ASSUMPTIONS ..... 6
4.1. Codes ..... 6
4.2. Geotechnical investigations ..... 7
4.3. Material Properties ..... 7
5. DESIGN LOADS ..... 7
5.1. Dead Load(DL) ..... 7
5.1.1. Roof Dead Loads ..... 7
5.1.2. Ground Floor Walls ..... 9
5.2. Live Load ..... 9
5.2.1. Roof Live Loads(Lr) ..... 9
5.3. Wind Load(WL) ..... 9
5.4. Wind Load (Analytical Verification) ..... 10
5.5. Seismic Load(E) ..... 14
5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads) ..... 14
6. DESIGN COMBINATIONS ..... 15
7. CRACKED SECTIONS ..... 16
8. DEFLECTIONS ..... 17
8.1. Maximum Allowable Deflection ..... 17
9. CONCRETE ROOF BEAM DESIGN ..... 18
9.1. Design of Ledger Beam(LB1) ..... 18
9.1.1. Staad Output For LB1 ..... 19
9.2. Design of Ledger Beam(LB2) ..... 21
9.2.1. Long Term Deflection Check For LB2 ..... 22



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2.2. Staad Output For LB2 ..... 26
9.3. Design of Ledger Beam(LB3) ..... 27
9.3.1. Staad Output For LB3 ..... 28
9.4. Design of Beam-(B1) ..... 29
9.4.1. Staad Output For B1 ..... 30
10. CONCRETE GRADE SLAB DESIGN ..... 31
11. DESIGN OF GRADE BEAM ..... 32
11.1. Grade Beam(GB1) ..... 32
11.1.1. Staad Output For GB1 ..... 33
12. CONCRETE COLUMN DESIGN ..... 34
12.1. Staad Output For Column CR & C1 ..... 35
13. CANTILEVER SLAB DESIGN ..... 36
14. FOUNDATION DESIGN ..... 38
References ..... 39
ATTACHMENT 1 STAAD INPUT DATA ..... 40
ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS ..... 46

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## DEFINITIONS AND ABBREVIATIONS

| | | |
|---|---|---|
| Document | : | Information and the medium on which it is contained (ISO 9000:2015) |
| EMPLOYER | : | 'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives |
| Contractor | : | Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract). |
| 'PMT' | : | Ma'aden Project Management Team |
| 'MMGP Project' | : | Mansourah Massarah Gold Processing Project |
| Attribute | : | Information element that describes a document or tag. |
| Supplier | : | An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor |
| Sub-Supplier | : | A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor. |
| QA | : | Quality Assurance |
| QC | : | Quality Control |
| Works | : | Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities |
| Typ. | : | Typical |
| HCS | : | Hollow Core Slab |
| Fc' | : | Concrete Compressive Strength |
| Fy | : | Yield Strength of Reinforcing Steel |
| $\phi$ | : | Strength Reduction Factor |
| SBC | : | Saudi Building Code |
| SASO | : | Saudi Standards, Metrology and Quality Organization |
| ACI | : | American Concrete Institute |
| ASCE | : | American Society of Civil Engineers |
| PCI | : | Precast/Prestressed Concrete Institute |


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Engineering Cum CCR Building.

## 2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

## 3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Engineering Office Cum CCR Building only.


PROPRIETARY & CONFIDENTIAL

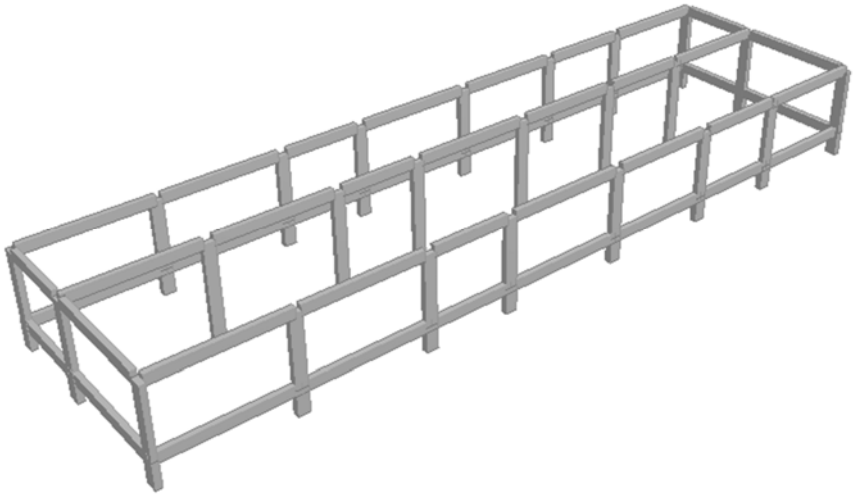
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

### 4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

### 4.3. Material Properties

**Compressive Strength of Concrete at 28th days**

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

**Yield Strength of Steel**

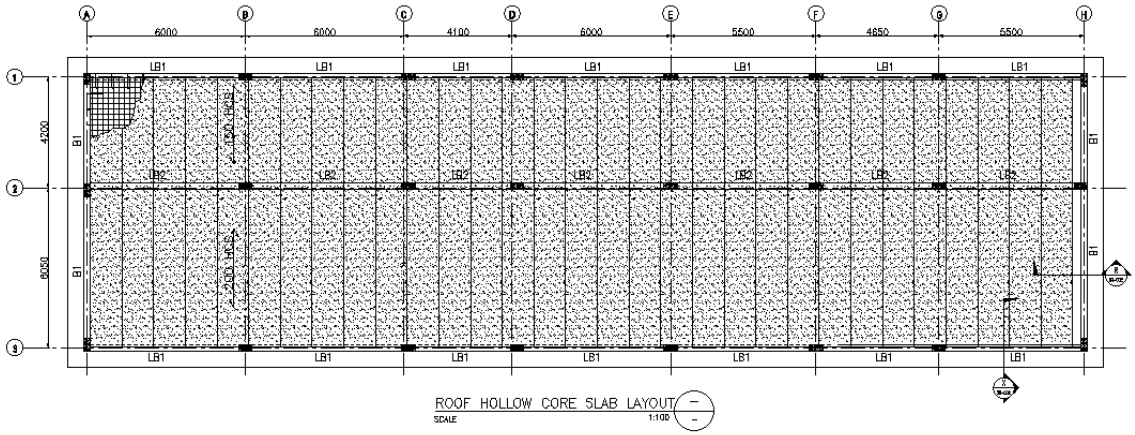
fy = 414 MPa (60 Ksi) for all reinforcing steel.

## 5. DESIGN LOADS

### 5.1. Dead Load(DL)

#### 5.1.1. Roof Dead Loads

| | |
|---|---|
| • Provision for Electrical Lighting | = 0.05 kN/m² |
| Ceiling | |
| Suspended steel channel system | = 0.10 kN/m² |
| Acoustical Fiber Tile (16mm) | = 0.05 kN/m² |
| Sub-total | = **0.2 kN/m²** |




PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

- Hollow Core Slab

(HCS-200)(To be confirmed by Vendor) b/w Grid Line's (2 & 3) = **3.10 KN/m2**

(HCS-150)(To be confirmed by Vendor) b/w Grid Line's (1 & 2) = **2.52 KN/m2**

| Prestressed Hollowcore Floor | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Overall Structural Depth (mm) | Self Wt- KN/M2 | No. of Cores in Unit | Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m² | | | | | | | |
| | | | 0.75 | 1.50 | 2.00 | 2.50 | 3.00 | 4.00 | 5.00 | 10.00 |
| | | | Clear Span in metres | | | | | | | |
| 100 | 2.38 | 0 | 5.39 | 5.19 | 5.08 | 4.91 | 4.72 | 4.40 | 4.14 | 3.30 |
| 150 | 2.52 | 11 | 7.40 | 7.40 | 7.40 | 7.40 | 7.28 | 6.80 | 6.40 | 5.12 |
| 150 (Dense | 2.94 | 11 | 7.40 | 7.40 | 7.40 | 7.35 | 7.08 | 6.64 | 6.27 | 5.06 |
| 200 | 3.10 | 11 | 9.40 | 9.40 | 9.40 | 9.40 | 9.26 | 8.69 | 8.21 | 6.65 |
| 250 | 3.43 | 6 | 10.80 | 10.80 | 10.80 | 10.80 | 10.80 | 10.80 | 10.56 | 8.60 |
| 300 | 3.88 | 6 | 13.50 | 13.50 | 13.50 | 13.50 | 13.18 | 12.44 | 11.81 | 9.70 |
| 350 | 4.23 | 6 | 15.00 | 15.00 | 15.00 | 15.00 | 14.89 | 14.08 | 13.39 | 11.06 |
| 400 | 4.57 | 6 | 16.00 | 16.00 | 16.00 | 16.00 | 15.97 | 15.13 | 14.42 | 11.98 |
| 450 | 4.92 | 6 | 18.00 | 18.00 | 18.00 | 17.70 | 17.19 | 16.33 | 15.58 | 13.02 |
| 500 | 5.27 | 6 | 18.00 | 18.00 | 18.00 | 18.00 | 18.00 | 17.13 | 16.38 | 13.76 |

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load b/w Grid's (2 & 3) =3.1+0.2+3.1= 6.40 **kN/m²**


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Total Dead Load b/w Grid's (1& 2) =2.52+0.2+3.1= 5.82 **kN/m²**

**TOTAL DL as UDL ON BEAMS**

- UDL on Ledger Beams Along Grid Line-1 = 4.20 x 5.82/2 = **12.2 Kn/m**
- UDL on Ledger Beams Along Grid Line-3 = 6.05 x 6.40/2 =**19.36 Kn/m**
- UDL on Ledger Beams Along Grid Line-2 =12.2+19.36 =**31.56 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

### 5.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

## 5.2. Live Load

### 5.2.1. Roof Live Loads(Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

**TOTAL Lr as UDL ON BEAMS**

- UDL on Ledger Beams Along Grid Line-1 = 4.20 x 0.50/2 = **1.05 Kn/m**
- UDL on Ledger Beams Along Grid Line-3 = 6.05 x 0.50/2 =**1.52 Kn/m**
- UDL on Ledger Beams Along Grid Line-2 =1.05+1.52 =**2.57 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

## 5.3. Wind Load(WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

PROPRIETARY & CONFIDENTIAL

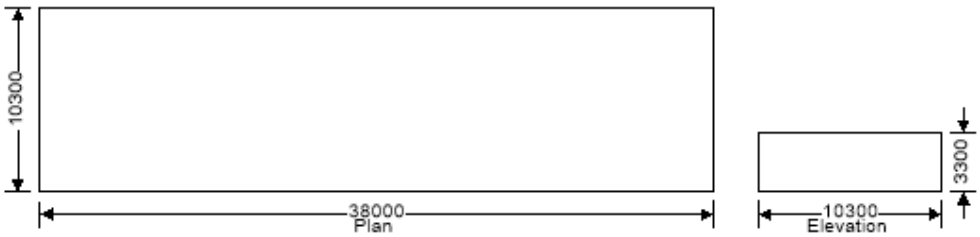
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 5.4. Wind Load (Analytical Verification)

**WIND LOADING (SBC-301)**

Using the directional design method




Building data

| | |
|---|---|
| Type of roof; | Flat |
| Length of building; | b = 38000 mm |
| Width of building; | d = 10300 mm |
| Height to eaves; | H = **3300** mm |
| Mean height; | h = **3300** mm |

General wind load requirements

| | |
|---|---|
| Basic wind speed; | V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001) |
| Risk category; | II(As per SBC-301, Table 1.6-1) |
| Velocity pressure exponent coeff ; | $K_d$ = **0.85**(As per SBC-301, Table 6.4-1) |
| Exposure category; | D(As per SBC-301, Table 6.4.2.3) |
| Enclosure classification; | Partially enclosed buildings(As per SBC-301, Section-6.2) |
| Internal pressure coef +ve; | $GC_{pi_p}$ = **0.55**(As per SBC-301, Figure-7.2-1) |
| Internal pressure coef –ve; | $GC_{pi_n}$ = **-0.55**(As per SBC-301, Figure-7.2-1) |
| Gust effect factor; | $G_f$ = **0.85**(As per SBC-301, Section-7.2.7.1) |
| Importance Factor; | 1(As per SBC-301, Section-9.5) |

Topography

| | |
|---|---|
| Topography factor not significant; | $K_{zt}$ = 1.0(As per SBC-301, Section-6.4.3.2) |
| Velocity pressure equation; | $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$ |


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

| z (mm) | $K_z$ (SBC-301,Table 7.2-2) | $q_z$ (kN/m$^2$) |
|---|---|---|
| 3300 | 1.03 | 1.29 |
| 3301 | 1.03 | 1.29 |

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, $GC_{pi}$ 0.55, - $c_{pe}$

| Zone | Ref. height (m) | Ext pressure coefficient $c_{pe}$ | Peak velocity pressure $q_p$ (kN/m$^2$) | Net pressure p (kN/m$^2$) | Area $A_{ref}$ (m$^2$) | Net force $F_w$ (kN) |
|---|---|---|---|---|---|---|
| A | 3.30 | 0.80 | 1.29 | 0.17 | 125.44 | 21.01 |
| B | 3.30 | -0.50 | 1.29 | -1.26 | 125.40 | -157.55 |
| C | 3.30 | -0.70 | 1.29 | -1.48 | 33.99 | -50.15 |
| D | 3.30 | -0.70 | 1.29 | -1.48 | 33.99 | -50.15 |

Walls load case 2 - Wind 0, $GC_{pi}$ -0.55, + $c_{pe}$

| Zone | Ref. height (m) | Ext pressure coefficient $c_{pe}$ | Peak velocity pressure $q_p$ (kN/m$^2$) | Net pressure p (kN/m$^2$) | Area $A_{ref}$ (m$^2$) | Net force $F_w$ (kN) |
|---|---|---|---|---|---|---|
| A | 3.30 | 0.80 | 1.29 | 1.58 | 125.44 | 198.81 |
| B | 3.30 | -0.50 | 1.29 | 0.16 | 125.40 | 20.20 |
| C | 3.30 | -0.70 | 1.29 | -0.06 | 33.99 | -1.97 |
| D | 3.30 | -0.70 | 1.29 | -0.06 | 33.99 | -1.97 |


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

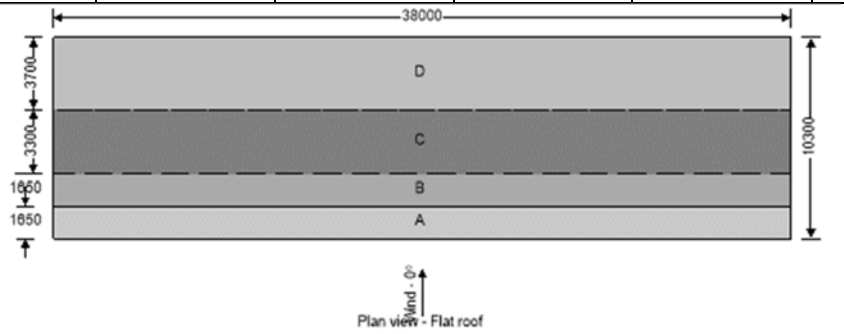
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Walls load case 3 - Wind 90, $GC_{pi}$ 0.55, - $c_{pe}$

| Zone | Ref. height (m) | Ext pressure coefficient $c_{pe}$ | Peak velocity pressure $q_p$ (kN/m²) | Net pressure p (kN/m²) | Area $A_{ref}$ (m²) | Net force $F_w$ (kN) |
|---|---|---|---|---|---|---|
| A | 3.30 | 0.80 | 1.29 | 0.17 | 34.00 | 5.70 |
| B | 3.30 | -0.22 | 1.29 | -0.94 | 33.99 | -32.11 |
| C | 3.30 | -0.70 | 1.29 | -1.48 | 125.40 | -185.02 |
| D | 3.30 | -0.70 | 1.29 | -1.48 | 125.40 | -185.02 |

Walls load case 4 - Wind 90, $GC_{pi}$ -0.55, + $c_{pe}$

| Zone | Ref. height (m) | Ext pressure coefficient $c_{pe}$ | Peak velocity pressure $q_p$ (kN/m²) | Net pressure p (kN/m²) | Area $A_{ref}$ (m²) | Net force $F_w$ (kN) |
|---|---|---|---|---|---|---|
| A | 3.30 | 0.80 | 1.29 | 1.58 | 34.00 | 53.89 |
| B | 3.30 | -0.22 | 1.29 | 0.47 | 33.99 | 16.07 |
| C | 3.30 | -0.70 | 1.29 | -0.06 | 125.40 | -7.27 |
| D | 3.30 | -0.70 | 1.29 | -0.06 | 125.40 | -7.27 |

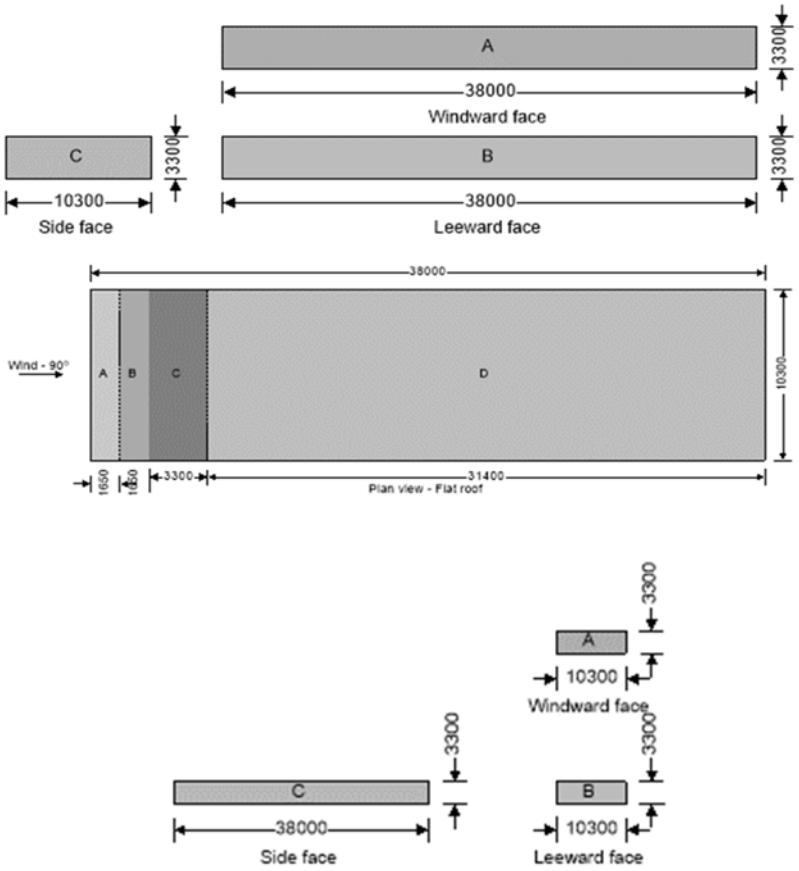




PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.




So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 5.5. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

$S_{D_1}$ = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

$S_{DS}$ = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where $S_{D_1}$ Damped Design Spectral Response Acceleration@1 Sec

$S_{DS}$ Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

## 5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

| | |
|---|---|
| For Beams: | I = 0.35Ig |
| For Columns: | I = 0.70Ig |
| For Walls: | |
| Un-cracked | I = 0.70Ig |
| Cracked | I = 0.35Ig |
| For Slabs | I = 0.25Ig |

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

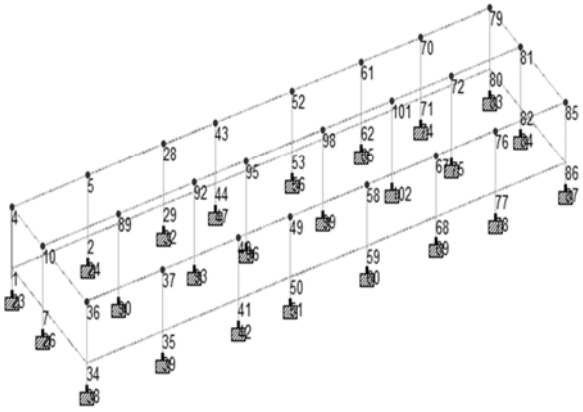
# 8. DEFLECTIONS

## 8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



ENG OFFICE-5th MAY-20.std - Node Displacement

| Node | L/C | X-Trans mm | Y-Trans mm | Z-Trans mm | Absolute mm | X-Rotan rad | Y-Rotan rad | Z-Rotan rad |
|---|---|---|---|---|---|---|---|---|
| 58 | 20 | 0.280 | -0.310 | 23.875 | 23.878 | 0.009 | 0.000 | 0.000 |
| 52 | 20 | 0.220 | -0.158 | -23.855 | 23.857 | -0.009 | -0.000 | 0.000 |
| 49 | 20 | 0.319 | -0.265 | 22.993 | 22.996 | 0.008 | -0.000 | -0.001 |
| 43 | 20 | 0.237 | -0.137 | -22.982 | 22.983 | -0.008 | 0.000 | -0.000 |
| 58 | 16 | 0.272 | -0.186 | 22.854 | 22.857 | 0.008 | 0.000 | 0.000 |
| 52 | 16 | 0.207 | -0.095 | -22.836 | 22.838 | -0.008 | -0.000 | 0.000 |
| 40 | 20 | 0.324 | -0.260 | 22.264 | 22.268 | 0.008 | -0.000 | 0.000 |
| 28 | 20 | 0.239 | -0.134 | -22.242 | 22.244 | -0.008 | 0.000 | 0.000 |
| 49 | 16 | 0.295 | -0.159 | 22.007 | 22.009 | 0.008 | -0.000 | -0.000 |
| 43 | 16 | 0.216 | -0.082 | -21.996 | 21.997 | -0.008 | 0.000 | -0.000 |
| 67 | 20 | 0.259 | -0.264 | 21.385 | 21.388 | 0.008 | 0.001 | 0.000 |
| 40 | 16 | 0.299 | -0.155 | 21.317 | 21.320 | 0.008 | -0.000 | 0.000 |
| 61 | 20 | 0.212 | -0.135 | -21.300 | 21.301 | -0.008 | -0.001 | 0.000 |
| 28 | 16 | 0.218 | -0.080 | -21.295 | 21.296 | -0.008 | 0.000 | 0.000 |
| 52 | 17 | 0.024 | -0.095 | 21.083 | 21.083 | 0.007 | 0.000 | 0.000 |
| 67 | 16 | 0.261 | -0.158 | 20.472 | 20.474 | 0.007 | 0.001 | 0.000 |
| 58 | 21 | 0.026 | -0.310 | 20.460 | 20.462 | 0.007 | 0.000 | 0.000 |

Maximum Deflection Attained is 24mm<<<33mm (H/100) .Hence Structure is safe against Drift Check.

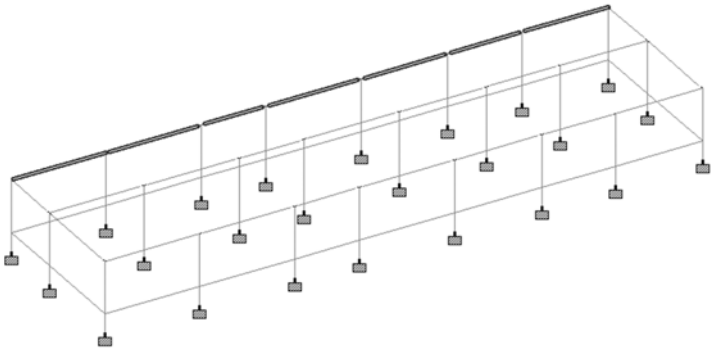

PROPRIETARY & CONFIDENTIAL

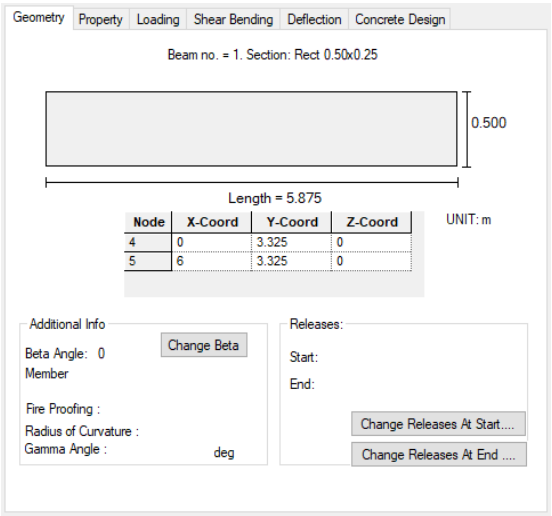
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

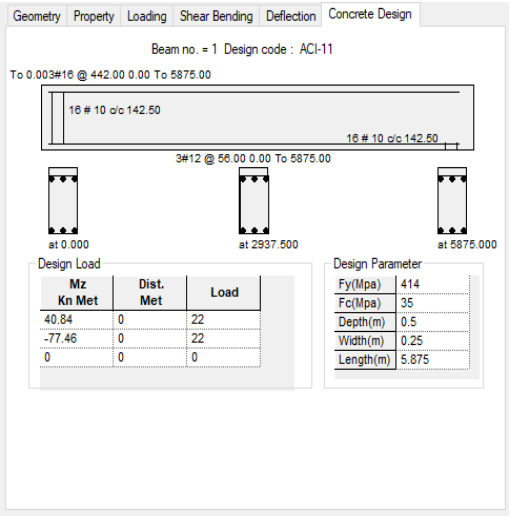
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

# 9. CONCRETE ROOF BEAM DESIGN

## 9.1. Design of Ledger Beam(LB1)






Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 77 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & $\phi$ =0.9

Ast1.req = **476 mm$^2$**


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **40 Knm (Sagging)**

Ast2.req = **240 mm²**

Hence Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bottom

Ast1.pro= **602mm² >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **602mm² >>>>>>(Ast.min, Ast2.req)**

### 9.1.1. Staad Output For LB1

```
        B E A M  N O.     1 D E S I G N  R E S U L T S - SHEAR

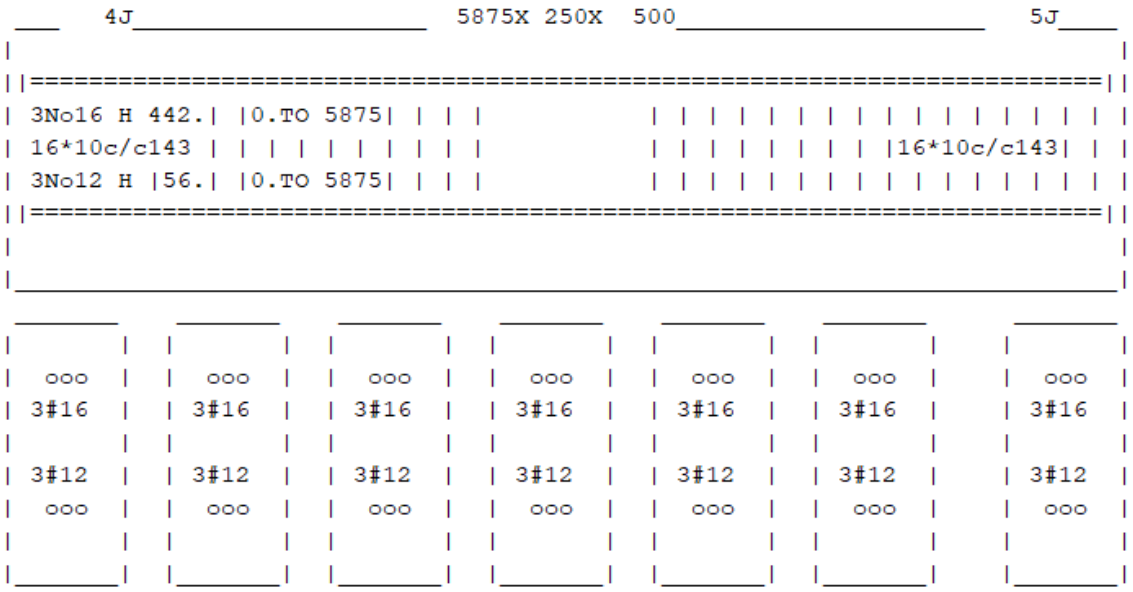

AT START SUPPORT - Vu=   29.42 KNS  Vc=  111.98 KNS  Vs=    0.00 KNS
Tu=   15.58 KN-MET  Tc=    4.0 KN-MET  Ts=   20.8 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1998. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   5.13 SQ.CM.

AT END   SUPPORT - Vu=   44.97 KNS  Vc=  110.76 KNS  Vs=    0.00 KNS
Tu=   12.72 KN-MET  Tc=    3.9 KN-MET  Ts=   17.0 KN-MET  LOAD    26
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1998. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.19 SQ.CM.
```


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



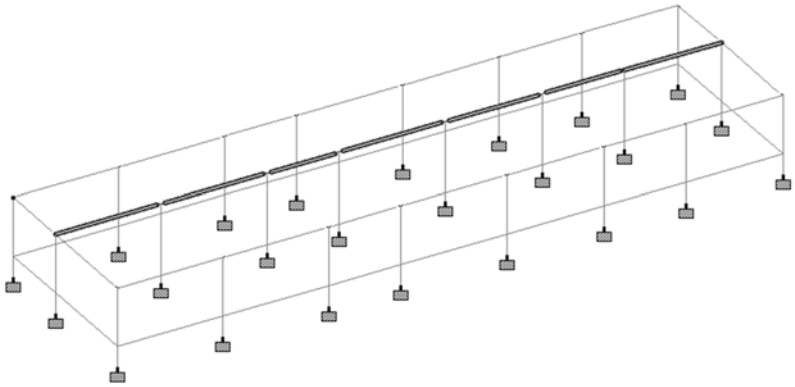


PROPRIETARY & CONFIDENTIAL

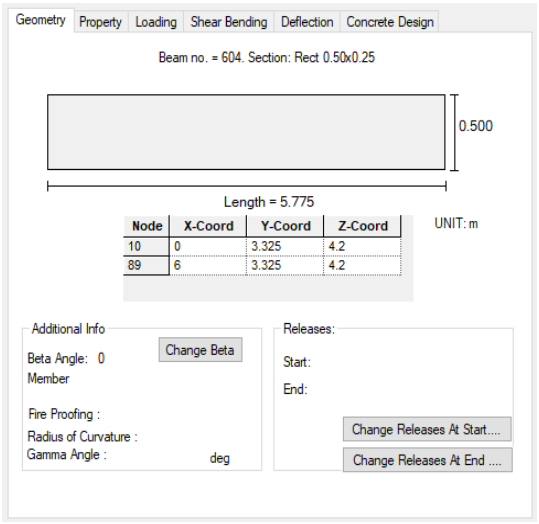
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

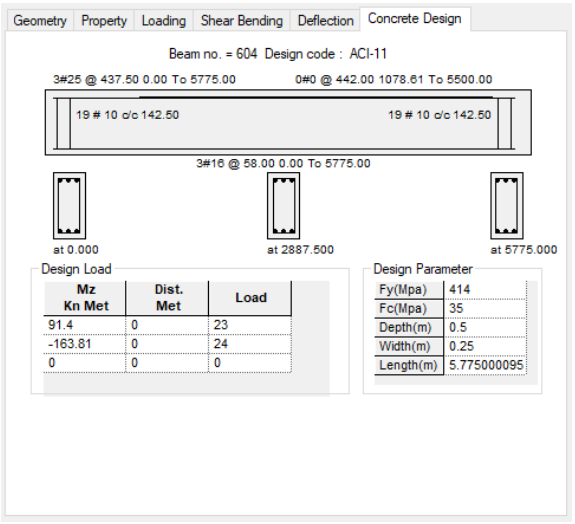
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 9.2. Design of Ledger Beam(LB2)



Design of Ledger Beam (LB2)




However Provide 3-25 dia-Full Straight Bars Top & 3-2 5
dia-Full Straight Bottom, for Beam LB2 considering Long term Deflection check in Section 9.2.1

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro **>>>>>>(Ast.min, Ast2.req)**


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

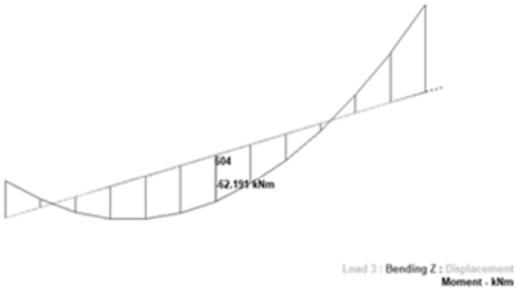
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 9.2.1. Long Term Deflection Check For LB2

**(LB2-BEAM-LONGTERM DEFLECTION CHECK)**

**INPUT DATA :**

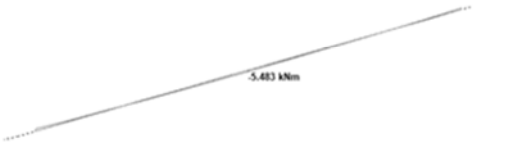
| | | | |
|---|---|---|---|
| Conc. Compressive stress, fc` | = | 35.00 | MPa |
| Reinforcement bar tensile stress, fy | = | 414.00 | MPa |
| Modulus Elasticity of concrete, | = | 27,806 | Mpa |
| Modulus Elasticity of steel, *Es* | = | 200,000 | Mpa |
| Unit wt. of concrete, Wconc | = | 24.00 | kN/m$^3$ |
| Unit wt. of steel | = | 77.00 | kN/m$^3$ |
| Depth of beam, h | = | 500 | mm |
| width of beam, b | = | 250 | mm |
| Length of beam, L | = | 6.00 | m |
| Bottom bar diam., db | = | 25.00 | mm |
| area of 1 bar Ab | = | 490.87 | mm$^2$ |
| no. of bottom bars | = | 3.00 | pcs. |
| Top bar diam., db | = | 25.00 | mm |
| area of 1 bar Ab | = | 490.87 | mm$^2$ |
| no. of top bars | = | 3.00 | pcs. |
| Stirrups diam. Dv | = | 10.00 | mm |
| area of 1 bar Av | = | 78.54 | mm$^2$ |
| concrete cover, cc | = | 40.00 | mm |
| $d_{eff.bot}$ (@ bottom), d | = | 437.50 | mm |
| (FROM STAAD RESULTS) Mdead | = | 65.00 | kN-m |
| Mlive | = | 6.00 | kN-m |




Page 1

**CHECK DEFLECTION:** $M_{d+l}$ = 71.00 kN-m

$M_{sus} = M_d + 0.50 M_l$ = 68.00 kN-m




***modulus of rupture, modulus of elasticity, modular ratio***

$f_r = 0.62\lambda\sqrt{f'_c}$ = 3.67 Mpa




$n_s = \frac{Es}{Ec}$ = 7.19 ←*ACI(318M-08) Sec. 9.5.2.3*

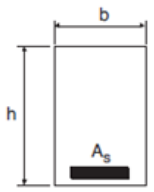
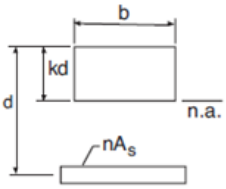
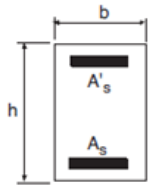
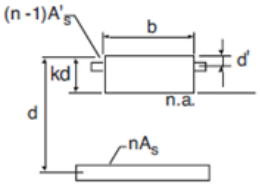
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

***gross and crack section of moment of inertia***

*Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section*

| Gross Section | Cracked Transformed Section | Gross and Cracked Moment of Inertia |
|---|---|---|
|  |  Without compression steel | $n = \frac{E_s}{E_c}$<br>$B = \frac{b}{(nA_s)}$<br>$I_g = \frac{bh^3}{12}$<br>Without compression steel<br>$kd = \left(\sqrt{2dB+1}-1\right)/B$<br>$I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$ |
|  |  With compression steel | With compression steel<br>$r = (n-1)A'_s/(nA_s)$<br>$kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$<br>$I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$ |

Page 2

$$I_g = \frac{bh^3}{12} = 2.60E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.02 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 154.78 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.16E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 2.25$$

***effective moment of inertia***

$$M_{cr} = \frac{ff_r I_g}{y_t} = 38.21 \text{ kN-m}$$ ←ACI(318M-08) Sec. 9.5.2.3

*under dead load*

$$\frac{M_{cr}}{M_d} = 0.59$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.45E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

*under sustained load*

$$\frac{M_{cr}}{M_{sus}} = 0.56$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.41E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

*under dead + live load*

$$\frac{M_{cr}}{M_{d+\lambda}} = 0.54$$

$$(I_e)_{d+\lambda} = \left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3\right] I_{cr} = 1.38E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

***initial or short-term deflection***

$$K = 1.00$$

$$(\Delta_i)_d = \frac{K(5/48)M_d\lambda^2}{Ec(I_e)_d} = 6.046 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\lambda^2}{Ec(I_e)_{sus}} = 6.492 \text{ mm}$$

$$(\Delta_i)_{d+\lambda} = \frac{K(5/48)M_{d+\lambda}\lambda^2}{Ec(I_e)_{d+\lambda}} = 6.932 \text{ mm}$$

$$(\Delta_i)_\lambda = (\Delta_i)_{d+\lambda} - (\Delta_i)_d = 0.885 \text{ mm}$$

*Table 10-3 Deflection Coeffcient K*

| | | K |
|---|---|---|
| 1. | Cantilevers (deflection due to rotation at supports not included) | 2.40 |
| 2. | Simple beams | 1.0 |
| 3. | Continuous beams | 1.2-0.2 $M_o/M_a$ |
| 4. | Fixed-hinged beams (midspan deflection) | 0.80 |
| 5. | Fixed-hinged beams (maximum deflection using maximum moment) | 0.74 |
| 6. | Fixed-fixed beams | 0.60 |

For other types of loading, K values are given in Ref. 10.2.

$M_o$ = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

$M_a$ = Net midspan moment.

**TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS**

| Type of member | Deflection to be considered | Deflection limitation |
|---|---|---|
| Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections | Immediate deflection due to live load *L* | $\ell/180^*$ |
| Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections | Immediate deflection due to live load *L* | $\ell/360$ |
| Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections | That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)† | $\ell/480^‡$ |
| Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections | | $\ell/240^§$ |

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

$$(\Delta_i)_\lambda \le \frac{\lambda}{180} = 33.33 \text{ mm} > 0.89 \text{ mm} \quad ok!$$

$$(\Delta_i)_\lambda \le \frac{\lambda}{360} = 16.67 \text{ mm} > 0.89 \text{ mm} \quad ok!$$

***additional long-term deflection***

$\xi = 2.00$

5 years or more .....2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$$\rho = \frac{As}{bd} = 0.0135$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.20 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.5$$

Page 4

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 7.76 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda = 8.65 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480} = 12.50 \text{ mm} > 8.65 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240} = 25.00 \text{ mm} > 8.65 \text{ mm} \quad ok!$$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

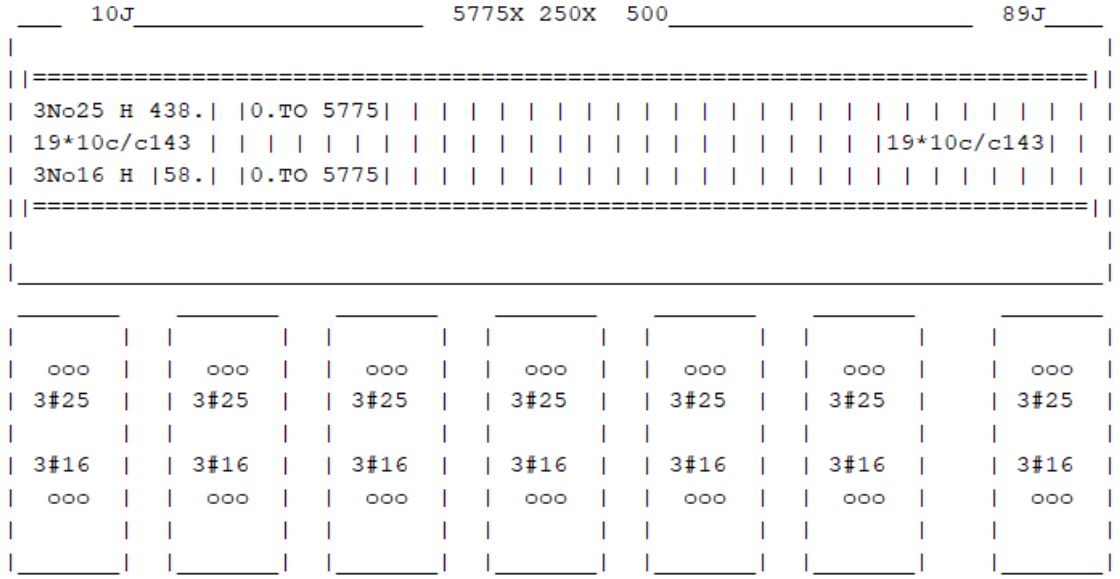
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 9.2.2. Staad Output For LB2

```
        B E A M  N O.   604 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   65.63 KNS  Vc=  133.87 KNS  Vs=    0.00 KNS
 Tu=    5.67 KN-MET  Tc=    3.9 KN-MET  Ts=    7.6 KN-MET  LOAD    33
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2448. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.87 SQ.CM.

 AT END   SUPPORT - Vu=  115.22 KNS  Vc=  109.44 KNS  Vs=   44.20 KNS
 Tu=    5.66 KN-MET  Tc=    4.0 KN-MET  Ts=    7.5 KN-MET  LOAD    30
 STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2448. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.87 SQ.CM.
```



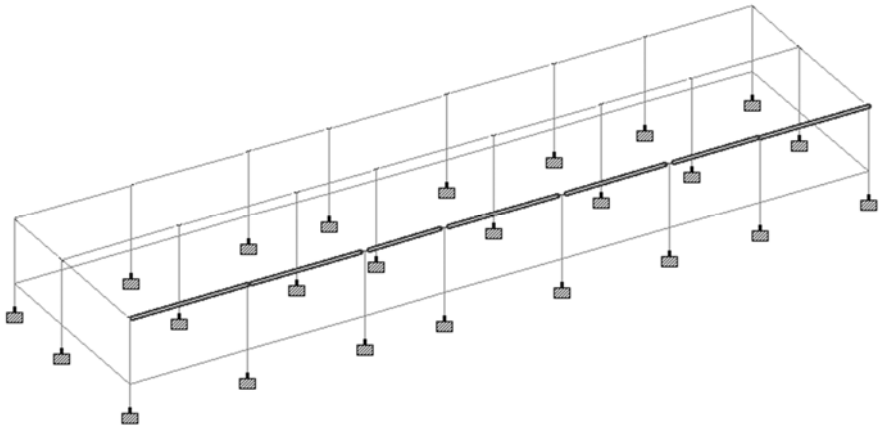


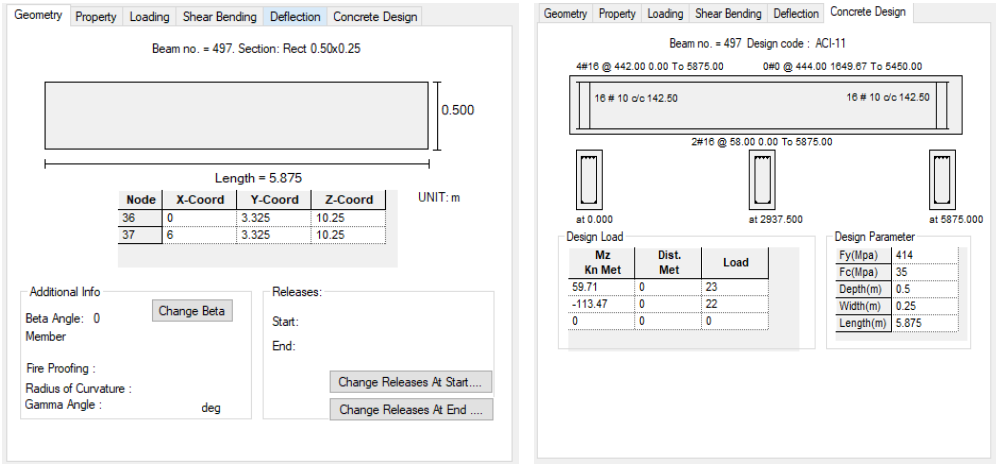
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 9.3. Design of Ledger Beam(LB3)






Design of Ledger Beam (LB3)

Hence Provide 3-20 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast.pro**>>>>>>(Ast.min, Ast.req)**


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

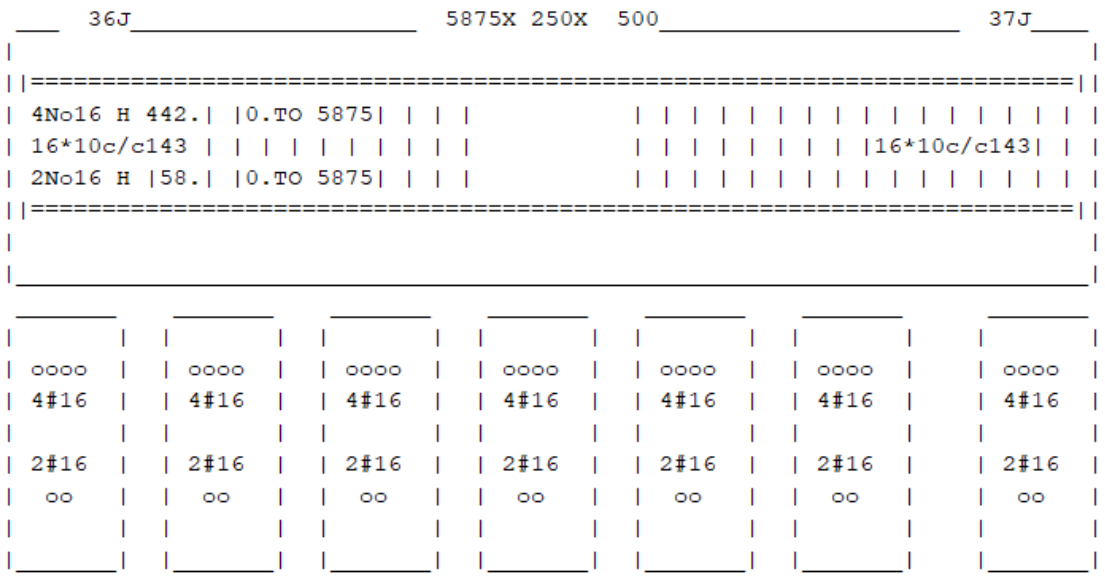
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 9.3.1. Staad Output For LB3

```
        B E A M  N O.   497 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   43.59 KNS  Vc=  113.91 KNS  Vs=    0.00 KNS
Tu=   15.66 KN-MET  Tc=    4.1 KN-MET  Ts=   20.9 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1998. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   5.16 SQ.CM.

AT END   SUPPORT - Vu=   48.58 KNS  Vc=  112.45 KNS  Vs=    0.00 KNS
Tu=   10.90 KN-MET  Tc=    4.0 KN-MET  Ts=   14.5 KN-MET  LOAD    31
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1998. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   3.59 SQ.CM.
```



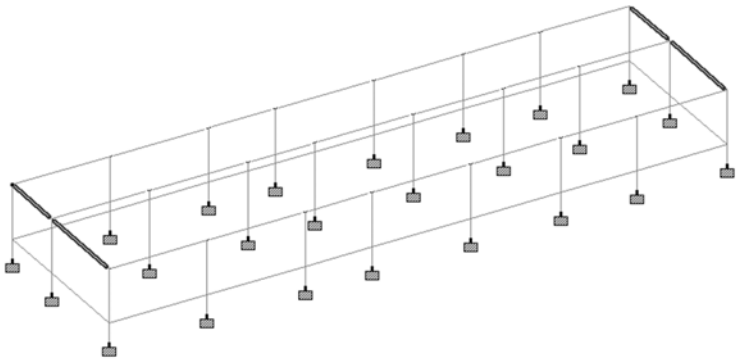


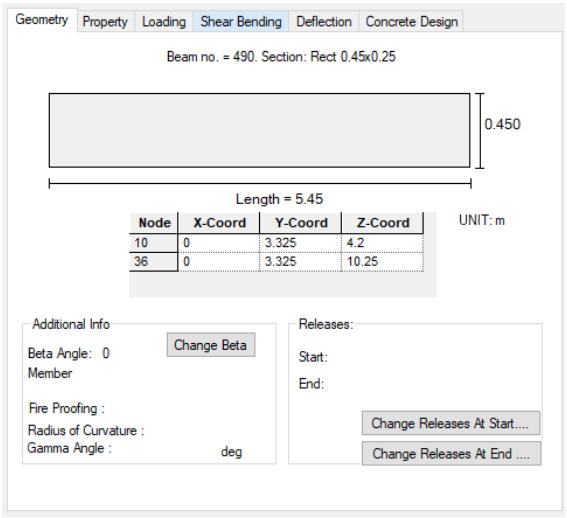
PROPRIETARY & CONFIDENTIAL

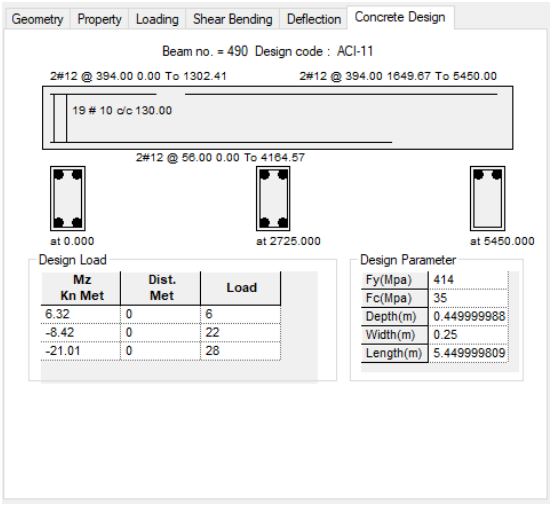
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 9.4. Design of Beam-(B1)






Provide 3-12 dia-Full Straight Bars Top & Bottom


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

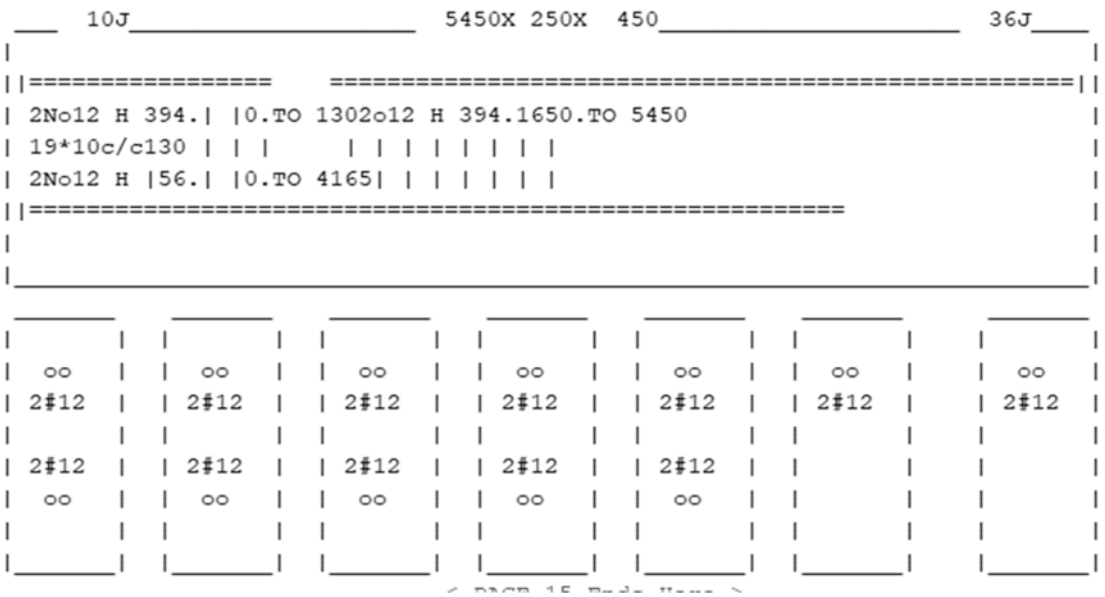
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 9.4.1. Staad Output For B1

```
                B E A M   N O.   490 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=     7.46 KNS   Vc=    93.84 KNS   Vs=     0.00 KNS
   Tu=     5.40 KN-MET   Tc=     3.4 KN-MET   Ts=     7.2 KN-MET   LOAD     23
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 130. MM  C/C FOR 2335. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.85 SQ.CM.

   AT END   SUPPORT - Vu=     6.42 KNS   Vc=    92.63 KNS   Vs=     0.00 KNS
   Tu=     1.23 KN-MET   Tc=     3.4 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
```





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

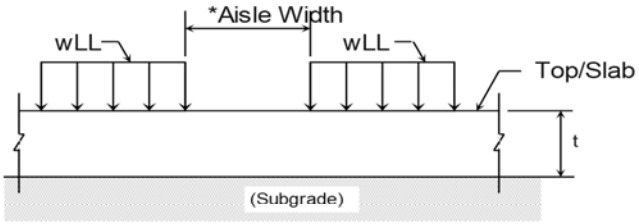
## 10. CONCRETE GRADE SLAB DESIGN

### CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

**Input Data:**

| | | | | | | |
|---|---|---|---|---|---|---|
| Slab Thickness, t = | 3.937 | in. | 100.000 | mm | | |
| Concrete Strength, f 'c = | 5000 | psi | 35.000 | kpa | | |
| Subgrade Modulus, k = | 39.7866 | pci | 10800 | Kn/m3 | (AS PER SECTION-7.7-SOIL REPORT) | |
| Factor of Safety, FS = | 1.500 | | 1.500 | | | |
| Uniform Live Load, wLL = | 104.45 | psf | 5.00 | kpa | | |




Concrete Slab on Grade with Uniform Loads

**Results:**

**Design Parameters:**

| | | | | | |
|---|---|---|---|---|---|
| Modulus of Rupture, MR = | 636.40 | psi | 4387.79 | kpa | MR = 9*SQRT(f 'c) |
| ow. Bending Stress, Fb = | 424.26 | psi | 2925.19 | kpa | Fb = MR/FS |
| Modulus of Elasticity, Ec = | 4030509 | psi | 27789329 | kpa | Ec = 57000*SQRT(f 'c) |
| Poisson's Ratio, μ = | 0.15 | | 0.15 | | μ = 0.15 (assumed for concrete) |
| Radius of Stiffness, Lr = | 26.94 | in. | 684.37 | mm | Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25 |
| Critical Aisle Width, Wcr = | 4.96 | ft. | 1511.77 | mm | Wcr = (2.209*Lr)/12 |

**Stationary Uniformly Distributed Live Loads:**

| | | | |
|---|---|---|---|
| wLL(allow) = | 682.05 | psf | wLL(allow) = 257.876*Fb*SQRT(k*t/Ec) |
| | 32.65 | kpa | **wLL(allow) >= wLL, O.K.** |

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

**Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area**

| Reinforcement type | $f_y$, psi | Minimum reinforcement ratio | |
|---|---|---|---|
| Deformed bars | < 60,000 | 0.0020 | |
| Deformed bars or welded wire reinforcement | ≥ 60,000 | Greater of: | $\frac{0.0018 \times 60,000}{f_y}$ |
| | | | 0.0014 |

Provide WWF-6mm Dia@200 c-c Both Ways-TOP

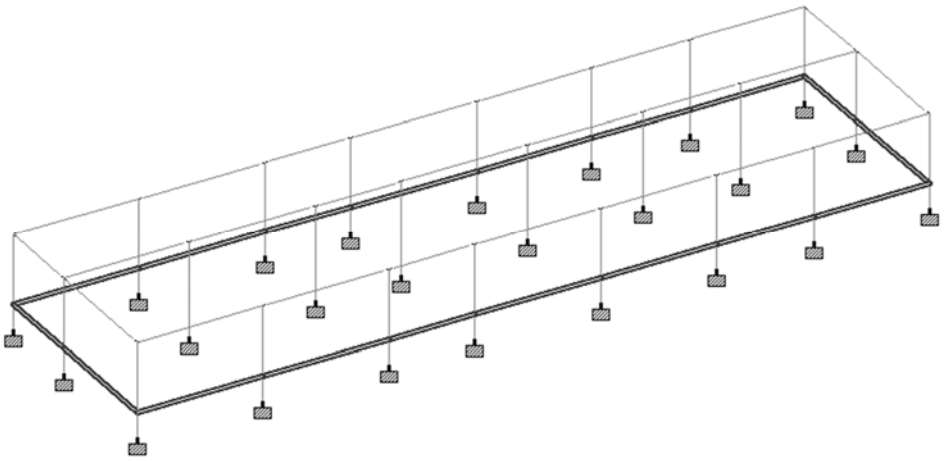

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

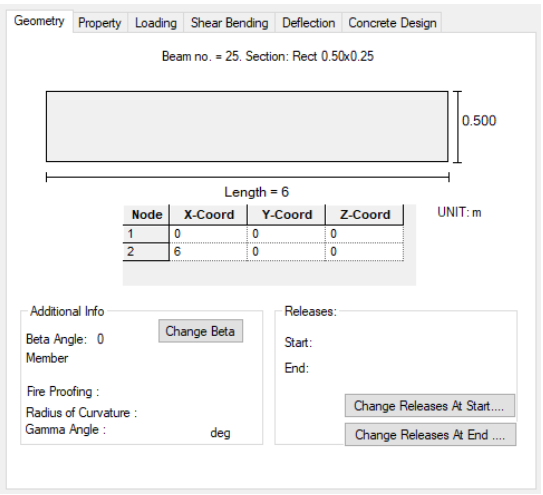
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

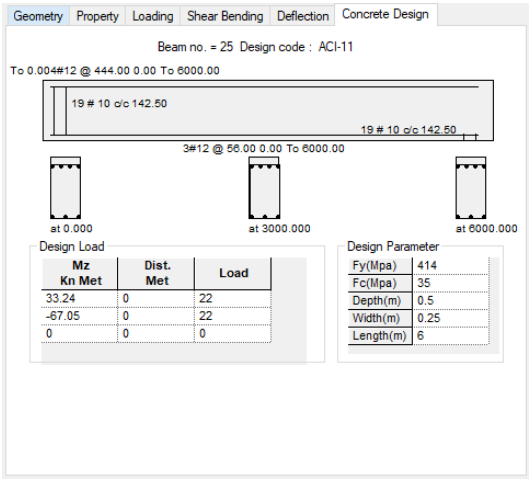
# 11. DESIGN OF GRADE BEAM

## 11.1. Grade Beam(GB1)



MOMENT ENVELOPE FOR BEAMS GB1




Provide minimum 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

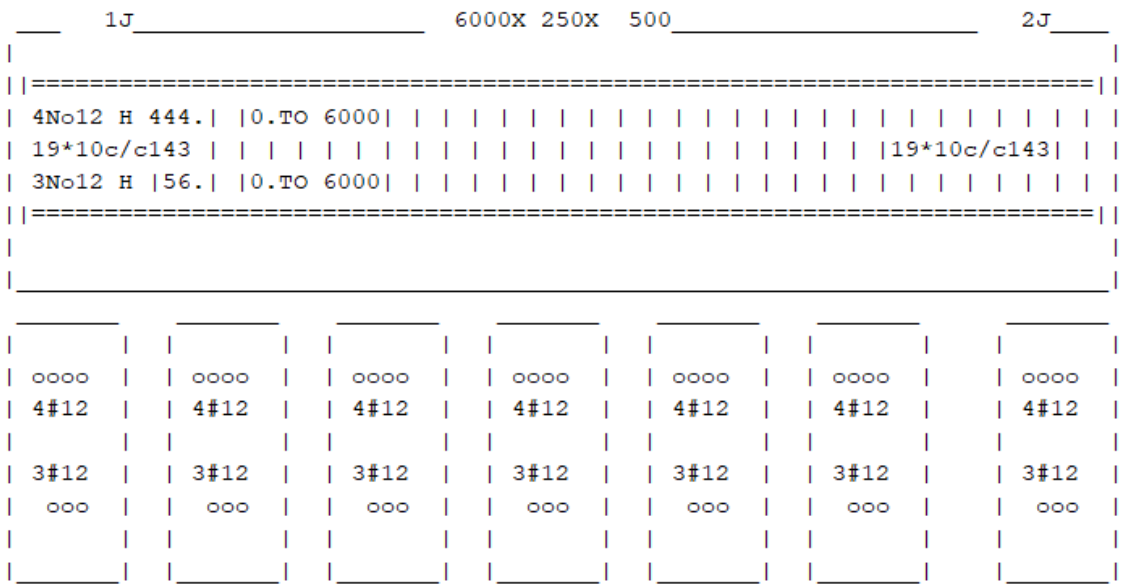
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

### 11.1.1. Staad Output For GB1

```
            B E A M   N O.     25 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=    43.69 KNS   Vc=  107.89 KNS   Vs=     0.00 KNS
   Tu=     4.92 KN-MET   Tc=     4.0 KN-MET   Ts=     6.6 KN-MET   LOAD     27
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2560. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.62 SQ.CM.

   AT END    SUPPORT - Vu=    48.18 KNS   Vc=  106.67 KNS   Vs=     0.00 KNS
   Tu=     4.92 KN-MET   Tc=     4.0 KN-MET   Ts=     6.6 KN-MET   LOAD     27
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2560. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.62 SQ.CM.
```



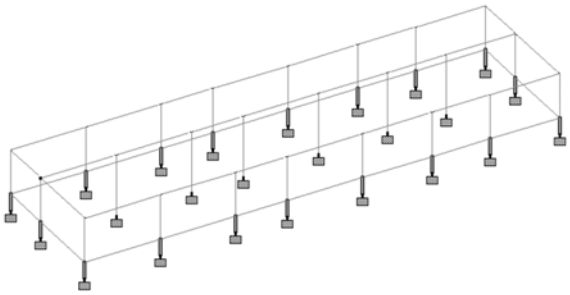


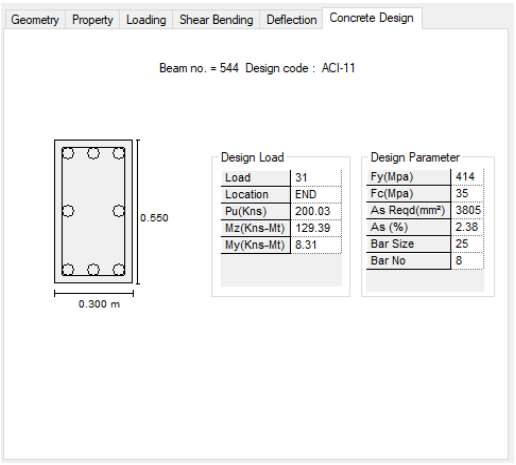
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

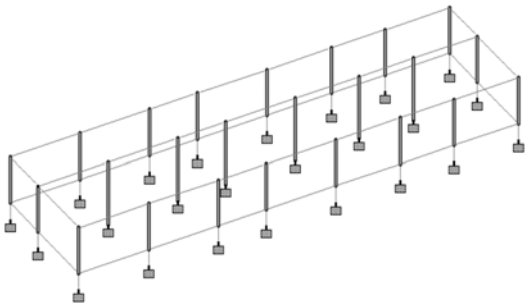
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

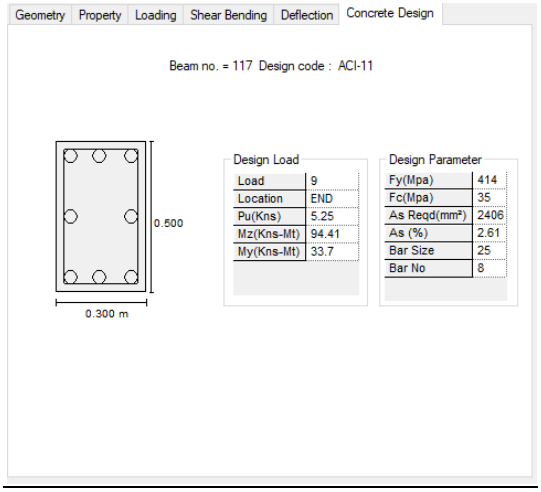
## 12. CONCRETE COLUMN DESIGN






Column's CR






Column's C1


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 12.1. Staad Output For Column CR & C1

```
        ACI 318-11   COLUMN  NO.   506   DESIGN RESULTS
        ================================================


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 550.0 X 300.0 MMS, TIED
         AREA OF STEEL REQUIRED =  3547.5  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

  12 - 20 MM               2.285       31      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 192.00 MM

        ACI 318-11   COLUMN  NO.   498   DESIGN RESULTS
        ================================================


-------------------------------< PAGE 55 Ends Here >--------------------------
  STAAD SPACE                                                  -- PAGE NO.   56

 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 550.0 X 250.0 MMS, TIED
         AREA OF STEEL REQUIRED =  2098.2  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

   8 - 20 MM               1.828       23      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 250.00 MM
```


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

| Geometrical Properties | | | Material Properties | | | |
|---|---|---|---|---|---|---|
| Span to center of support "$sp_1$" | 0.8 | met | Weight of reinforced concrete "$\gamma_r$ | 24 | | kN/m³ |
| Clear Span | 0.6 | met | Strength of concrete "$f_c$" | 35 | | N/mm² |
| Clear span + depth of slab "$sp_2$" | 0.75 | met | Strength of steel "$f_y$" | 420 | | N/mm² |
| Max. eff span le = min ($sp_1$;$sp_2$) | 0.8 | met | Clear cover to concrete("c") | 20 | | mm |
| Width of Slab "b" | 1000 | mm | Flexure strength reduction factor | 0.9 | | |
| Thickness of slab "t" | 150 | mm | Shear strength reduction factor | 0.75 | | |
| Eff. Depth "d" | 125 | | Modulus of Elasticity of steel "Es" | 2E+05 | | N/mm² |
| **Loading** | | | Modulus of Elasticity of concrete | 27806 | | N/mm² |
| Self Weight of slab | 3.6 | kN/m² | Modular Ratio "n = Es/Ec" | 7.19 | | |
| Floor Finish | 3.5 | kN/m² | **Provided Reinforcement** | | | |
| Wall load | 0 | kN/m² | Top layer Nos. | 6 | Φ 10 | mm |
| Total dead Load "$U_{dl}$" | 7.1 | kN/m² | Bottom layer Nos. | 0 | Φ 12 | mm |
| Live load | 1 | kN/m² | Area of Top steel "$A_{s\ t}$" | 468 | | mm² |
| Total Ultimate Load "$Uu_{tot}$"(1.2D+1.6L) | 10.12 | kN/m² | Area of Bottom steel "$A_{s\ b}$" | 0 | | mm² |
| Moment due to Overhang Point Load(0.125*0.15*24*0.6) | 0.27 | kNm | "db" = 10 mm | | | |
| **Calculations** | | | | | | |
| **Shear force Vu = U*le** | **8.096** | **kN** | | | | |
| **Neg. Moment Mu = U*le²/2** | **3.51** | **kNm** | | | | |

**Check for Section capacity**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Depth of comp. block "a" = $A_{s\ t}*f_y/(0.85*f'_c*b)$ | 6.61 | mm | | | | | | |
| Section capacity $\Phi Mn = \Phi*A_{s\ t}*f_y*(d-a/2)$ | **21.94** | **kNm** | **O.K** | **>** | **3.51** | **kNm** | (Mu) | |

Shear requirements

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Shear capacity $\Phi Vc = \Phi*0.17*\sqrt{f'c}*b*d$ | **94.29** | **kN** | **O.K** | **>** | **8.096** | **kN** | (Vu) | |

**Serviceability requirements**

| | | |
|---|---|---|
| $d' = c+d_b/2$ | 26 | mm |
| $B = b/(n*A_{s\ t})$ | 0.297 | |
| $r = (n-1)*A_{s\ b}/(n*A_{s\ t})$ | 0 | |
| $k_d = (\sqrt{(2*d*B(1+r*d'/d)+(1+r)^2)}-(1+r))/B$ | 29.01 | mm |
| $I_{cr} = bkd^3/3+n*A_{s\ t}*(d-k_d)^2+(n-1)*A_{s\ b}*(k_d-d'$ | 3.915E+07 | mm⁴ |
| **Calculate Gross moment of inertia** | | |
| Area of slab A = t*b | 2E+05 | mm² |
| Depth of neutral axis "y" | 75 | mm |
| Distance of top fiber from N.A "$y_t$" | 75 | mm |
| Gross moment of inertia "$I_g$" | 2.813E+08 | mm⁴ |
| Allowable stress $fr = 0.62\sqrt{f'c}$ | 3.67 | mm² |
| Cracking moment $M_{cr} = fr*I_g/y_t$ | 13.75 | kNm |


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

**Service moments**

| | | |
|---|---|---|
| Moment due to dead load "$M_{dl}$" | 2.54 | kNm |
| Moment due to live load "Mll" | 0.32 | kNm |
| Total Moment Ms = $M_{dl}+M_{ll}$ | 2.86 | kNm |

**Deflection due to dead + live loads**

| | | | |
|---|---|---|---|
| R(dl+ll) = $M_{cr}/Ms$ | 1.000 | | |
| $I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$ | 2.813E+08 | $mm^4$ | |
| $\delta = K*5/48*Ms*le^2/(Ec*I_e)$ | 0.08 | mm | **K = 3.2 (conservative)** |

**Deflection due to dead load**

| | | |
|---|---|---|
| R(dl) = $M_{cr}/Md$ | 5.411 | |
| $I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$ | 3.839E+10 | $mm^4$ |
| $\delta = K*5/48*Md*le^2/(Ec*I_e)$ | 0.00 | mm |

**Deflection due to live load**

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

**Long term deflection**

| | | |
|---|---|---|
| % of liveload sustained "SL" | 50 | % |
| $M_{sus} = M_{dl}+SL*M_{ll}$ | 2.70 | kNm |
| R(sus) = $M_{cr}/Msus$ | 5.091 | |
| $I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$ | 3.198E+10 | |
| $\delta = K*5/48*Msus*le^2/(Ec*I_e)$ | 0.00 | mm |

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

$\xi$ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = $\lambda_\Delta$*sus deflection + ll deflection 0.079 mm

**Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K**


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## 14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

**TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES**

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

| Profile / Borehole | Foundation Type | Depth of Foundation below FGL | Footing Size (B x L) | Net Allowable Bearing Capacity Based on | | Recommended Net Allowable Bearing Capacity | Estimated Total Settlement |
|---|---|---|---|---|---|---|---|
| | | | | Settlement | Shear Failure | | |
| - | - | (m) | (m) | (kN/m$^2$) | (kN/m$^2$) | (kN/m$^2$) | (mm) |
| BH-01 | Square Isolated | 1.20 | 1.0 x 1.0 | 1500 | 570 | 570 | 25 |
| | | 1.50 | | 2000 | 850 | 850 | 25 |
| | | 1.20 | 2.0 x 2.0 | 1000 | 750 | 750 | 25 |
| | | 1.50 | | 1200 | 980 | 980 | 25 |
| | | 1.20 | 3.0 x 3.0 | 775 | 900 | 775 | 25 |
| | | 1.50 | | 950 | 1120 | 950 | 25 |
| | | 1.20 | 4.0 x 4.0 | 625 | 1040 | 625 | 25 |
| | | 1.50 | | 750 | 1250 | 750 | 25 |
| | | 1.20 | 5.0 x 5.0 | 525 | 1172 | 525 | 25 |
| | | 1.50 | | 600 | 1342 | 600 | 25 |

Refer Attachment- II for Foundation Design calculations


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

## REFERENCES

| | | |
|---|---|---|
| 1.) | BUILDING PLANS | MD-216-4000-EG-AR-GAD-1075 |
| 2.) | OVERALL LAYOUT PLAN | MD-216-4000-EG-GAD-1074 |
| 3.) | LAYOUT OF FDN. COL. & GRADE BEAM | MD-216-4000-EG-CV-GAD-1078 |
| 4.) | DETAIL OF FDN. COL. & GRADE BEAM | MD-216-4000-EG-CV-DED-1079 |
| 5.) | LAYOUT OF ROOF & GRADE SLAB | MD-216-4000-EG-CV-GAD-1087 |
| 6.) | DETAIL OF ROOF & GRADE SLAB | MD-216-4000-EG-CV-DED-1088 |
| 7.) | GEOTECHNICAL REPORT | MD-216-3000-EG-CV-RPT-1008 |
| 8.) | STRUCTURAL DESIGN CRITERIA | MD-216-0000-EG-ST-CRT-0001 |


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

# ATTACHMENT 1 STAAD INPUT DATA


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
STAAD STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 6 0 0; 4 0 3.325 0; 5 6 3.325 0; 7 0 0 4.2; 10 0 3.325 4.2;
23 0 -1.2 0; 24 6 -1.2 0; 26 0 -1.2 4.2; 28 12 3.325 0; 29 12 0 0;
32 12 -1.2 0; 34 0 0 10.25; 35 6 0 10.25; 36 0 3.325 10.25; 37 6 3.325 10.25;
38 0 -1.2 10.25; 39 6 -1.2 10.25; 40 12 3.325 10.25; 41 12 0 10.25;
42 12 -1.2 10.25; 43 16.1 3.325 0; 44 16.1 0 0; 47 16.1 -1.2 0;
49 16.1 3.325 10.25; 50 16.1 0 10.25; 51 16.1 -1.2 10.25; 52 22.1 3.325 0;
53 22.1 0 0; 56 22.1 -1.2 0; 58 22.1 3.325 10.25; 59 22.1 0 10.25;
60 22.1 -1.2 10.25; 61 27.6 3.325 0; 62 27.6 0 0; 65 27.6 -1.2 0;
67 27.6 3.325 10.25; 68 27.6 0 10.25; 69 27.6 -1.2 10.25; 70 32.25 3.325 0;
71 32.25 0 0; 72 32.25 3.325 4.2; 74 32.25 -1.2 0; 75 32.25 -1.2 4.2;
76 32.25 3.325 10.25; 77 32.25 0 10.25; 78 32.25 -1.2 10.25; 79 37.75 3.325 0;
80 37.75 0 0; 81 37.75 3.325 4.2; 82 37.75 0 4.2; 83 37.75 -1.2 0;
84 37.75 -1.2 4.2; 85 37.75 3.325 10.25; 86 37.75 0 10.25; 87 37.75 -1.2 10.25;
89 6 3.325 4.2; 90 6 -1.2 4.2; 92 12 3.325 4.2; 93 12 -1.2 4.2;
95 16.1 3.325 4.2; 96 16.1 -1.2 4.2; 98 22.1 3.325 4.2; 99 22.1 -1.2 4.2;
101 27.6 3.325 4.2; 102 27.6 -1.2 4.2;
MEMBER INCIDENCES
1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;
34 7 26; 478 5 28; 479 29 28; 483 2 29; 485 29 32; 488 7 34; 490 10 36;
497 36 37; 498 34 36; 499 35 37; 500 35 34; 501 34 38; 502 35 39; 503 37 40;
504 41 40; 505 41 35; 506 41 42; 507 28 43; 508 29 44; 513 40 49; 514 41 50;
516 44 43; 519 44 47; 524 50 49; 525 50 51; 526 43 52; 527 44 53; 532 49 58;
533 50 59; 535 53 52; 538 53 56; 543 59 58; 544 59 60; 545 52 61; 546 53 62;
551 58 67; 552 59 68; 554 62 61; 557 62 65; 562 68 67; 563 68 69; 564 61 70;
565 62 71; 570 67 76; 571 68 77; 573 71 70; 574 72 75; 575 79 81; 576 71 74;
579 81 85; 581 77 76; 582 77 78; 583 70 79; 584 71 80; 585 72 81; 587 76 85;
588 77 86; 589 80 79; 590 82 81; 591 80 83; 592 82 84; 593 86 85; 594 86 87;
595 89 90; 597 92 93; 599 95 96; 601 98 99; 603 101 102; 604 10 89; 605 89 92;
606 92 95; 607 95 98; 608 98 101; 609 101 72; 610 80 82; 611 82 86;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
3 4 8 479 498 499 504 516 524 535 543 554 562 573 574 581 589 590 593 595 -
597 599 601 603 PRIS YD 0.25 ZD 0.55
1 25 30 478 483 488 497 500 503 505 507 508 513 514 526 527 532 533 545 546 -
551 552 564 565 570 571 583 TO 585 587 588 604 TO 611 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
31 32 34 485 501 502 506 519 525 538 544 557 563 576 582 591 592 -
594 PRIS YD 0.3 ZD 0.55
MEMBER PROPERTY AMERICAN
11 490 575 579 PRIS YD 0.45 ZD 0.25
CONSTANTS
BETA 90 MEMB 4 32 479 485 499 502 504 506 516 519 524 525 535 538 543 544 -
554 557 562 563 573 574 576 581 582 595 597 599 601 603
```

```
MATERIAL CONCRETE ALL
SUPPORTS
23 24 26 32 38 39 42 47 51 56 60 65 69 74 75 78 83 84 87 90 93 96 99 102 FIXED
******************************************************************************
MEMBER OFFSET
1 497 START 0.125 0 0
564 570 END -0.125 0 0
478 503 507 513 526 532 545 551 564 570 583 605 TO 609 START 0.225 0 0
478 503 507 513 526 532 545 551 604 TO 609 END -0.225 0 0
11 490 575 579 START 0 0 0.3
11 490 575 579 END 0 0 -0.3
MEMBER CRACKED
1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 527 532 533 -
545 546 551 552 564 565 570 571 575 579 583 TO 585 587 588 604 TO 610 -
611 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 525 535 538 543 -
544 554 557 562 563 573 574 576 581 582 589 TO 595 597 599 601 -
603 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
DEFINE IBC 2003
SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
25 30 483 488 500 505 508 514 527 533 546 552 565 571 584 588 610 611 UNI 12
1 478 507 526 545 564 583 UNI 12.2
497 503 513 532 551 570 587 UNI 19.36
585 604 TO 609 UNI 31.56
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 40.000 M 15.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.887 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.63495 1.63495 HEIG  0  4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 40.000 M 15.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.887 -0.267 0.550
!> END GENERATED DATA BLOCK
INT -1.02105 -1.02105 HEIG  0  3.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 40.000 M 15.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.887 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.51984 -1.51984 HEIG  0  3.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 11.000 M 40.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.61078 1.61078 HEIG  0  4.572
```

```
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 11.000 M 40.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.27453 -1.27453 HEIG  0  3.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 11.000 M 40.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.4987 -1.4987 HEIG  0  3.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 30 483 488 500 505 508 514 527 533 546 552 565 571 584 588 610 -
611 UNI GY -12
1 478 507 526 545 564 583 UNI GY -12.2
11 490 UMOM GZ 1
575 579 UMOM GZ -1
1 478 507 526 545 564 583 UMOM GX -1
497 503 513 532 551 570 587 UMOM GX 1
497 503 513 532 551 570 587 UNI GY -19.36
585 604 TO 609 UNI GY -31.56
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
1 478 507 526 545 564 583 UNI GY -1.05
11 490 UMOM GZ 0.1
575 579 UMOM GZ -0.1
1 478 507 526 545 564 583 UMOM GX -0.1
497 503 513 532 551 570 587 UMOM GX 0.1
497 503 513 532 551 570 587 UNI GY -1.52
585 604 TO 609 UNI GY -2.57
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 3.4
WIND LOAD -X 1 TYPE 2 YR 0 3.4
WIND LOAD -Z 1 TYPE 3 YR 0 3.4
WIND LOAD -Z -1 TYPE 3 YR 0 3.4
LOAD 6 LOADTYPE Wind  TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 3.4
WIND LOAD -Z 1 TYPE 5 YR 0 3.4
WIND LOAD -X 1 TYPE 6 YR 0 3.4
WIND LOAD -X -1 TYPE 6 YR 0 3.4
LOAD 7 LOADTYPE Temperature  TITLE TEMP
TEMPERATURE LOAD
1 3 4 8 11 478 479 490 497 TO 499 503 504 507 513 516 524 526 532 535 543 -
545 551 554 562 564 570 573 575 579 581 583 587 589 590 593 TEMP 25
```

LOAD 77 LOADTYPE Wind  TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4
WIND LOAD -Z -1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD COMB 8 SLS: (DL)
3 1.0
LOAD COMB 9 SLS: (DL+T)
3 1.0 7 1.0
LOAD COMB 10 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 11 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 12 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 13 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 14 SLS: (D+0.75WX+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 16 SLS: (0.6D+WX)
3 0.6 5 1.0
LOAD COMB 17 SLS: (0.6D+WZ)
3 0.6 6 1.0
LOAD COMB 18 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 19 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 20 SLS: (D+WX)
3 1.0 5 1.0
LOAD COMB 21 SLS: (D+WZ)
3 1.0 6 1.0
LOAD COMB 22 ULS: (1.4 DL)
3 1.4
LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 29 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 30 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 31 ULS: (0.90D+1.6WX)
3 0.9 5 1.6
LOAD COMB 32 ULS: (0.90D+1.6WZ)
3 0.9 6 1.6
LOAD COMB 33 ULS: (0.90D+EZ)
3 0.9 1 1.0

```
LOAD COMB 34 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 35 SLS: (DL+W-z)
3 1.0 77 1.0
LOAD COMB 36 SLS: 0.6DL+Wz-
3 0.6 77 1.0
LOAD COMB 37 ULS: (0.9DL+1.6Wz-ve)
3 0.9 77 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MINSEC 10 ALL
SFACE 0.5 MEMB 1 478 497 503 507 513 526 532 545 551 564 570
EFACE 0.5 MEMB 1 478 497 503 507 513 526 532 545 551 564 570
DESIGN BEAM 1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 -
527 532 533 545 546 551 552 564 565 570 571 575 579 583 TO 585 587 588 604 -
605 TO 611
DESIGN COLUMN 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 525 -
535 538 543 544 554 557 562 563 573 574 576 581 582 589 TO 595 597 599 601 -
603
END CONCRETE DESIGN
FINISH
```

# ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

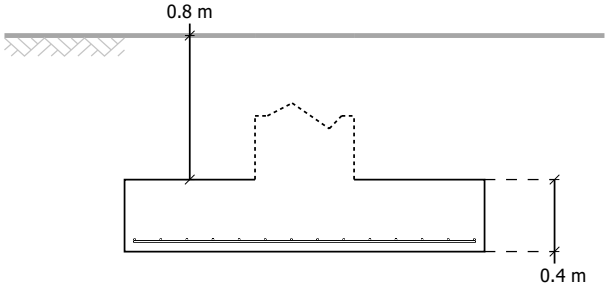
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

# Isolated Footing Design(ACI 318-08)

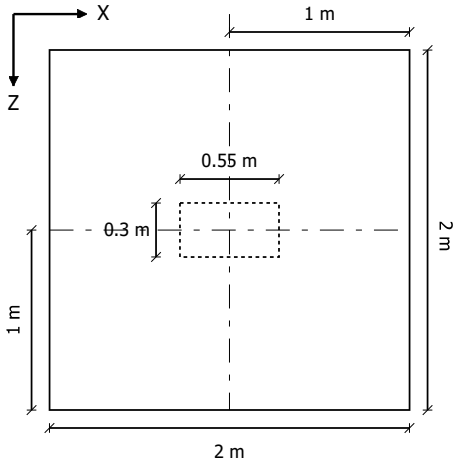
| Footing No. | Group ID | Foundation Geometry | | |
|---|---|---|---|---|
| - | - | Length | Width | Thickness |
| 74 | 1 | 2.000m | 2.000m | 0.400m |

| Footing No. | Footing Reinforcement | | | | Pedestal Reinforcement | |
|---|---|---|---|---|---|---|
| - | Bottom Reinforcement($M_z$) | Bottom Reinforcement($M_x$) | Top Reinforcement($M_z$) | Top Reinforcement($M_x$) | Main Steel | Trans Steel |
| 74 | 14 - ϕ12 | 14 - ϕ12 | N/A | N/A | N/A | N/A |

FOUNDATION-F1




Elevation




Plan

## Input Values

### Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2000.000mm
Footing Width - Z (Fw) : 2000.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

### Column Dimensions

Column Shape : Rectangular
Column Length - X ($D_{col}$) : 0.550m

Column Width - Z ($B_{col}$) : 0.300m

## Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

# Design Parameters

## Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

## Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 500.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

## Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

## Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

---

# Design Calculations

## Footing Size

Initial Length ($L_o$) = 2.000m

Initial Width ($W_o$) = 2.000m

## Load Combinations

| Load Combination/s- Service Stress Level | | | | |
|---|---|---|---|---|
| Load Combination Number | Load Combination Title | Load Combination Factor | Soil Bearing Factor | Self Weight Factor |
| 9 | SLS: (DL+T) | 1.00 | 1.00 | 1.00 |
| 10 | SLS: (D+LR) | 1.00 | 1.00 | 1.00 |
| 11 | SLS: (D+0.75LR) | 1.00 | 1.00 | 1.00 |
| 12 | SLS: (D+0.7EZ) | 1.00 | 1.00 | 1.00 |
| 13 | SLS: (D+0.7EX) | 1.00 | 1.00 | 1.00 |
| 14 | SLS: (D+0.75WX+0.75LR) | 1.00 | 1.00 | 1.00 |
| 15 | SLS: (D+0.75WZ+0.75LR) | 1.00 | 1.00 | 1.00 |
| 16 | SLS: (0.6D+WX) | 1.00 | 1.00 | 1.00 |
| 17 | SLS: (0.6D+WZ) | 1.00 | 1.00 | 1.00 |
| 18 | SLS: (0.6D+0.7EX) | 1.00 | 1.00 | 1.00 |
| 19 | SLS: (0.6D+0.7EZ) | 1.00 | 1.00 | 1.00 |
| 20 | SLS: (D+WX) | 1.00 | 1.00 | 1.00 |
| 21 | SLS: (D+WZ) | 1.00 | 1.00 | 1.00 |
| 35 | SLS: (DL+W-Z) | 1.00 | 1.00 | 1.00 |
| 36 | SLS: 0.6DL+WZ- | 1.00 | 1.00 | 1.00 |
| 77 | WL-Z | 1.00 | 1.00 | 1.00 |

| Load Combination/s- Strength Level | | | | |
|---|---|---|---|---|
| Load Combination Number | Load Combination Title | Load Combination Factor | Soil Bearing Factor | Self Weight Factor |
| 22 | ULS: (1.4 DL) | 1.00 | 1.00 | 1.00 |
| 23 | ULS: (1.4D+1.4T+0.5LR) | 1.00 | 1.00 | 1.00 |
| 24 | ULS: (1.4D+1.4T+0.5LR) | 1.00 | 1.00 | 1.00 |
| 25 | ULS: (1.2D+0.80WX+1.6LR) | 1.00 | 1.00 | 1.00 |
| 26 | ULS: (1.2D+1.6WZ+0.5LR) | 1.00 | 1.00 | 1.00 |
| 27 | ULS: (1.2D+1.6WX+0.5LR) | 1.00 | 1.00 | 1.00 |
| 28 | ULS: (1.2D+1.6WZ+0.5LR) | 1.00 | 1.00 | 1.00 |
| 29 | ULS: (1.2D+EX) | 1.00 | 1.00 | 1.00 |
| 30 | ULS: (1.2D+EZ) | 1.00 | 1.00 | 1.00 |
| 31 | ULS: (0.90D+1.6WX) | 1.00 | 1.00 | 1.00 |
| 32 | ULS: (0.90D+1.6WZ) | 1.00 | 1.00 | 1.00 |
| 33 | ULS: (0.90D+EZ) | 1.00 | 1.00 | 1.00 |
| 34 | ULS: (0.90D+EX) | 1.00 | 1.00 | 1.00 |
| 37 | ULS: (0.9DL+1.6WZ-VE) | 1.00 | 1.00 | 1.00 |
| 77 | WL-Z | 1.00 | 1.00 | 1.00 |

## Applied Loads

| Applied Loads - Service Stress Level | | | | | |
|---|---|---|---|---|---|
| LC | Axial (kN) | Shear X (kN) | Shear Z (kN) | Moment X (kNm) | Moment Z (kNm) |
| 9 | 171.034 | -10.855 | 0.109 | -3.233 | -2.485 |
| 10 | 175.770 | -8.504 | 0.436 | -2.473 | 4.973 |
| 11 | 174.424 | -8.658 | 0.427 | -2.415 | 5.037 |
| 12 | 170.378 | -9.134 | 3.439 | 8.113 | 5.267 |
| 13 | 171.301 | -4.645 | 0.515 | -1.908 | -2.135 |
| 14 | 174.715 | -6.427 | -15.933 | -42.081 | 1.978 |
| 15 | 174.410 | -7.406 | 18.095 | 40.887 | 4.001 |
| 16 | 102.620 | -2.496 | -21.575 | -54.234 | -0.941 |
| 17 | 102.214 | -3.802 | 23.796 | 56.391 | 1.756 |
| 18 | 103.147 | -0.998 | 0.356 | -1.010 | -4.226 |
| 19 | 102.223 | -5.487 | 3.280 | 9.010 | 3.175 |
| 20 | 170.775 | -6.144 | -21.416 | -55.131 | 1.150 |
| 21 | 170.369 | -7.449 | 23.954 | 55.494 | 3.847 |
| 35 | 170.387 | -5.719 | -18.264 | -48.134 | 2.333 |
| 36 | 102.232 | -2.071 | -18.422 | -47.237 | 0.242 |
| 77 | -0.000 | 3.399 | -18.661 | -45.891 | -2.895 |

| Applied Loads - Strength Level | | | | | |
|---|---|---|---|---|---|
| LC | Axial (kN) | Shear X (kN) | Shear Z (kN) | Moment X (kNm) | Moment Z (kNm) |
| 22 | 238.542 | -12.765 | 0.556 | -3.140 | 7.319 |
| 23 | 242.138 | -14.891 | 0.172 | -4.642 | -3.607 |
| 24 | 242.138 | -14.891 | 0.172 | -4.642 | -3.607 |
| 25 | 213.387 | -7.580 | -16.911 | -45.370 | 2.604 |
| 26 | 207.127 | -7.965 | 38.188 | 89.572 | 3.937 |
| 27 | 207.776 | -5.876 | -34.405 | -87.427 | -0.379 |
| 28 | 207.127 | -7.965 | 38.188 | 89.572 | 3.937 |
| 29 | 205.771 | -4.552 | 0.644 | -2.212 | -4.245 |
| 30 | 204.452 | -10.965 | 4.822 | 12.102 | 6.329 |
| 31 | 153.969 | -3.447 | -34.544 | -86.639 | -1.820 |
| 32 | 153.319 | -5.536 | 38.049 | 90.360 | 2.496 |
| 33 | 153.336 | -8.230 | 4.703 | 12.775 | 4.760 |
| 34 | 154.655 | -1.817 | 0.525 | -1.540 | -5.813 |
| 37 | 153.348 | -2.767 | -29.500 | -75.445 | 0.073 |
| 77 | -0.000 | 3.399 | -18.661 | -45.891 | -2.895 |

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.000m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.920m^2$

**Note: $A_{min}$ is an initial estimation.**
**P = Critical Factored Axial Load(without self weight/buoyancy/soil).**
**$q_{max}$ = Respective Factored Bearing Capacity.**

## Final Footing Size

Length ($L_2$) = 2.000 m Governing Load Case : # 9

Width ($W_2$) = 2.000 m Governing Load Case : # 9

Depth ($D_2$) = 0.400 m Governing Load Case : # 26

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area ($A_2$) = 4.000 $m^2$

Final Soil Height = 0.800 m

Footing Self Weight = 38.399 kN

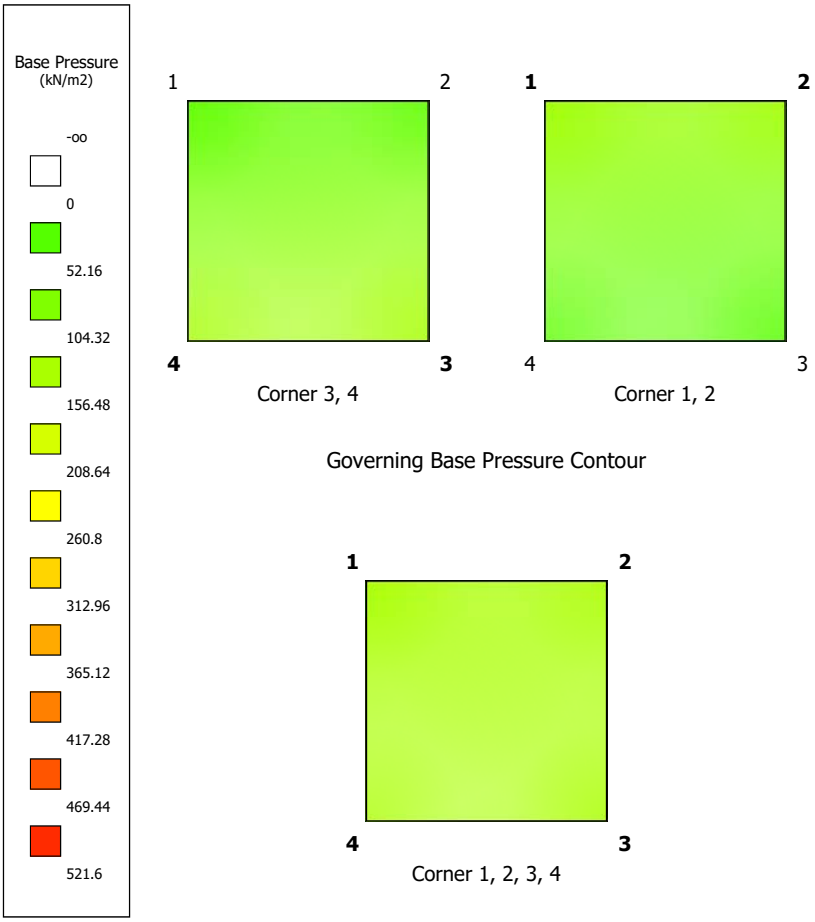
Gross Soil Bearing Capacity = 521.60 kN/m2

Soil Weight On Top Of Footing = 55.222 kN

## Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



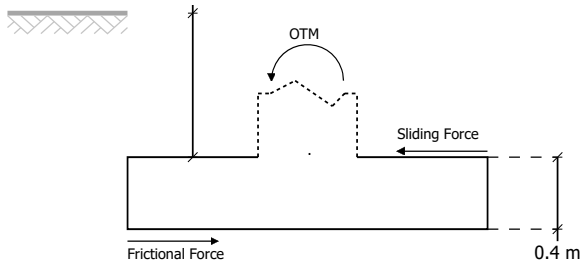

| Load Case | Pressure at corner 1 ($q_1$) (kN/m2) | Pressure at corner 2 ($q_2$) (kN/m2) | Pressure at corner 3 ($q_3$) (kN/m2) | Pressure at corner 4 ($q_4$) (kN/m2) | Area of footing in uplift ($A_u$) ($m^2$) |
|---|---|---|---|---|---|
| 20 | **116.5782** | 111.1670 | 15.6195 | 21.0307 | 0.000 |
| 20 | 116.5782 | **111.1670** | 15.6195 | 21.0307 | 0.000 |
| 21 | 22.3103 | 12.0696 | **109.6844** | 119.9252 | 0.000 |
| 21 | 22.3103 | 12.0696 | 109.6844 | **119.9252** | 0.000 |

If $A_u$ is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

| Load Case | Pressure at corner 1 ($q_1$) (kN/m2) | Pressure at corner 2 ($q_2$) (kN/m2) | Pressure at corner 3 ($q_3$) (kN/m2) | Pressure at corner 4 ($q_4$) (kN/m2) |
|---|---|---|---|---|
| 20 | **116.5782** | 111.1670 | 15.6195 | 21.0307 |
| 20 | 116.5782 | **111.1670** | 15.6195 | 21.0307 |
| 21 | 22.3103 | 12.0696 | **109.6844** | 119.9252 |
| 21 | 22.3103 | 12.0696 | 109.6844 | **119.9252** |

Stability Check




| - | Factor of safety against sliding | | | Factor of safety against overturning | | | |
|---|---|---|---|---|---|---|---|
| Load Case No. | Along X-Direction | Along Z-Direction | Resultant | Required FOS | About X-Direction | About Z-Direction | Required FOS |
| 9 | 12.922 | 1287.029 | 18.273 | 1.500 | 83.511 | 142.655 | 1.500 |
| 10 | 16.789 | 327.116 | 23.712 | 1.500 | 117.963 | 32.189 | 1.500 |
| 11 | 16.409 | 333.014 | 23.178 | 1.500 | 120.178 | 31.558 | 1.500 |
| 12 | 15.318 | 40.689 | 20.274 | 1.500 | 28.006 | 29.617 | 1.500 |
| 13 | 30.226 | 272.853 | 42.487 | 1.500 | 156.692 | 957.841 | 1.500 |
| 14 | 22.129 | 8.926 | 11.707 | 1.500 | 5.573 | 59.028 | 1.500 |
| 15 | 19.181 | 7.851 | 10.275 | 1.500 | 5.605 | 38.516 | 1.500 |
| 16 | 41.664 | 4.821 | 6.772 | 1.500 | 3.149 | 3426.481 | 1.500 |
| 17 | 27.298 | 4.362 | 6.091 | 1.500 | 2.997 | 59.820 | 1.500 |
| 18 | 104.486 | 293.105 | 139.186 | 1.500 | 228.648 | 51.468 | 1.500 |
| 19 | 18.916 | 31.646 | 22.962 | 1.500 | 19.141 | 36.504 | 1.500 |
| 20 | 22.809 | 6.543 | 8.895 | 1.500 | 4.178 | 73.346 | 1.500 |
| 21 | 18.782 | 5.841 | 7.888 | 1.500 | 4.083 | 38.696 | 1.500 |
| 35 | 24.468 | 7.661 | 10.340 | 1.500 | 4.793 | 57.182 | 1.500 |
| 36 | 50.111 | 5.635 | 7.919 | 1.500 | 3.618 | 183.172 | 1.500 |
| 77 | 14.597 | 2.659 | 3.700 | 1.500 | 1.787 | 22.050 | 1.500 |

## Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 9
Governing Disturbing Force : -10.855kN
Governing Restoring Force : 140.267kN
Minimum Sliding Ratio for the Critical Load Case : 12.922
Critical Load Case for Overturning about X-Direction : 77
Governing Overturning Moment : -53.355kNm
Governing Resisting Moment : 95.347kNm
Minimum Overturning Ratio for the Critical Load Case : 1.787

## Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 77
Governing Disturbing Force : -18.661kN
Governing Restoring Force : 49.619kN
Minimum Sliding Ratio for the Critical Load Case : 2.659
Critical Load Case for Overturning about Z-Direction : 77
Governing Overturning Moment : -4.255kNm
Governing Resisting Moment : 93.817kNm

Minimum Overturning Ratio for the Critical Load Case : 22.050

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 77

Governing Disturbing Force : 18.968kN

Governing Restoring Force : 70.172kN

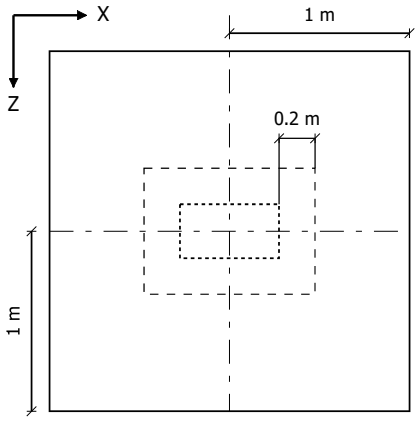
Minimum Sliding Ratio for the Critical Load Case : 3.700

## Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

## Shear Calculation

### Punching Shear Check




Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, $d_{eff}$ = D - $C_{cover}$ - 0.5 * $d_b$ = 0.344m

For rectangular column, $\beta_c$ = $B_{col}$ / $D_{col}$ = 1.833

Effective depth, $d_{eff}$, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 287.432kN, Load Case # 23

From ACI Cl.11.11.2, $b_o$ for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.076m

Equation 11-31, $V_{c1}$ = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2173.720kN

Equation 11-32, $V_{c2}$ = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3364.859kN

Equation 11-33, $V_{c3}$ = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2079.210kN

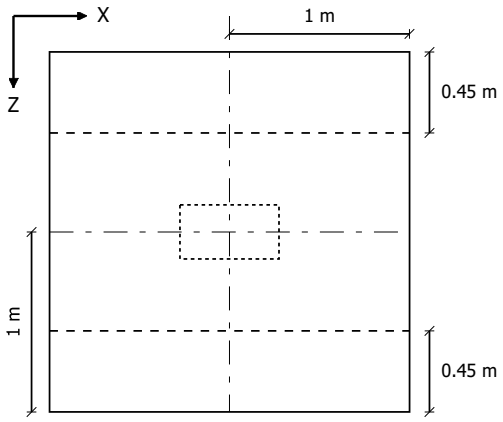
Punching shear strength, $V_c$ = 0.75 X minimum of ($V_{c1}$, $V_{c2}$, $V_{c3}$) = 1559.408kN

0.75 X $V_c$ > $V_u$ hence, OK

## One-Way Shear Check

### One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)




Plan

From ACI Cl.11.2.1.1, $V_c$ = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 675.946kN

Distance along X to design for shear, $D_x$ = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.506m

Check that 0.75 X $V_c$ > $V_{ux}$ where $V_{ux}$ is the shear force for the critical load cases at a distance $d_{eff}$ from the face of the column caused by bending about the X axis.

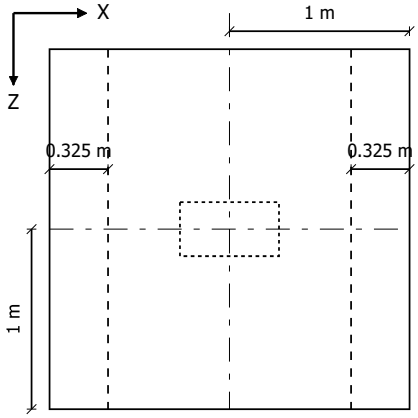
From above calculations, 0.75 X $V_c$ = 506.960 kN

Critical load case for $V_{ux}$ is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 135.286 kN

0.75 X $V_c$ > $V_{ux}$ hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)




Plan

From ACI Cl.11.2.1.1, $V_c$ =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 675.946 kN

Distance along X to design for shear, $D_z$ = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.381 m

Check that **0.75 X $V_c$ > $V_{uz}$** where $V_{uz}$ is the shear force for the critical load cases at a distance $d_{eff}$ from the face of the column caused by bending about the Z axis.

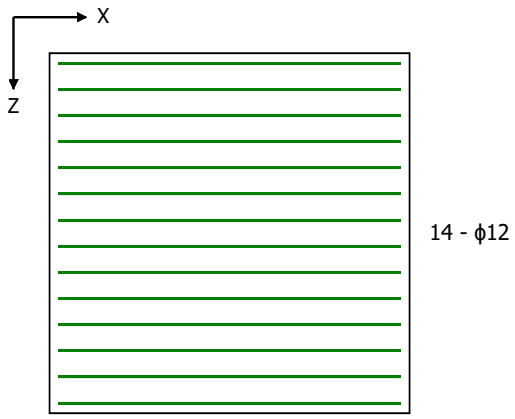
From above calculations, 0.75 X $V_c$ = 506.960 kN

Critical load case for $V_{uz}$ is # 22 $V_{uz} = V_{uz}|_{z=D_z}$ = 69.025 kN

0.75 X $V_c$ > $V_{uz}$ hence, OK

Flexure About Z-Axis

## Design For Bottom Reinforcement Parallel to X Axis




Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 22

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

| | | |
|---|---|---|
| Effective Depth $d_{eff}$ = | | 0.344 m |
| Factor $\beta_1$ from ACI Cl.10.2.7.3 = | | 0.796 |
| From ACI Cl. 10.3.2, $\rho_{bal}$ = | $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ | 0.03317 |
| From ACI Cl. 10.3.3, $\rho_{max}$ = | $0.75 \times \rho_{bal}$ = | 0.02488 |
| From ACI Cl. 7.12.2, $\rho_{min}$ = | | 0.00160 |
| From Ref. 1, Eq. 3.8.4a, constant m = | $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = | 14.118 |

**Calculate reinforcement ratio ρ for critical load case**

| | | |
|---|---|---|
| Design for flexure about Z axis is performed at the face of the column at a distance, $D_x$ = | $0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ = | 0.725 m |
| Ultimate moment, | $M_u\big\vert_{z=D_z}$ = | 47.361 kNm |
| Nominal moment capacity, $M_n$ = | $\frac{M_u}{\Phi}$ = | 52.623 kNm |
| (Based on effective depth) Required ρ = | $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = | 0.00053 |
| | (Based on gross depth) ρ x $d_{eff}$ / Depth = | 0.00046 |
| Since | ρ= $\rho_{min}$ | $\rho_{min}$ Governs |
| Area of Steel Required, $A_s$ = | $\rho \times W \times d_{eff}$ = | 1279.167 mm2 |

Selected bar Size = #12

Minimum spacing allowed ($S_{min}$) = = 100.000mm

Selected spacing (S) = 150.000mm

$S_{min}$<= S <= $S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

**Based on spacing reinforcement increment; provided reinforcement is**

**#12 @ 150mm o.c.**

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, $D_L$ = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar}$ = $\frac{A_s}{A_{bar}}$ = 14

**Because the number of bars is rounded up, make sure new reinforcement ratio < $\rho_{max}$**

Total reinforcement area, $A_{s_total}$ = $N_{bar}$ X (Area of one bar) = 1583.352 mm2

$d_{eff}$ = D - $C_{cover}$ - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho$ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00230

From ACI Cl.7.6.1, minimum req'd clear distance between bars

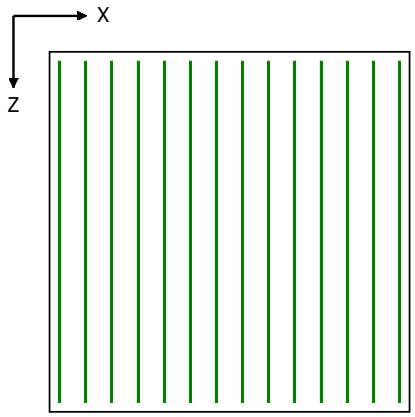
$C_d$ = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.238

**Check to see if width is sufficient to accomodate bars**

## Flexure About X-Axis

### Design For Bottom Reinforcement Parallel to Z Axis




14 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth $d_{eff}$ = 0.332 m

Factor $\beta_1$ from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal}$ = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max}$ = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, $\rho_{min}$ = 0.00166

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

**Calculate reinforcement ratio ρ for critical load case**

| | | | |
|---|---|---|---|
| Design for flexure about X axis is performed at the face of the column at a distance, $D_z$ = | $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = | 1.150 | m |
| Ultimate moment, | $M_u\big\vert_{x=D_x}$ = | 95.359 | kNm |
| Nominal moment capacity, $M_n$ = | $\frac{M_u}{\Phi}$ = | 105.955 | kNm |
| (Based on effective depth) Required ρ = | $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^{\ 2}\right)}}\right]$ = | 0.00115 | |
| | (Based on gross depth) ρ x $d_{eff}$ / Depth = | 0.00096 | |
| Since | ρ= $\rho_{min}$ | $\rho_{min}$ Governs | |
| Area of Steel Required, $A_s$ = | $\rho \times W \times d_{eff}$ = | 1330.930 | mm2 |

Selected Bar Size = #12

Minimum spacing allowed ($S_{min}$) = 100.000mm

Selected spacing (S) = 150.000mm

$S_{min}$<= S <= $S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

**Based on spacing reinforcement increment; provided reinforcement is**

**#12 @ 150mm o.c.**

| | | | |
|---|---|---|---|
| Required development length for bars = | $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = | =0.410 | m |
| Available development length for bars, $D_L$ = | $0.5 \times \left(L - D_{col}\right) \cdot - C_{cover}$ = | 0.800 | m |
| Try bar size # 12 | Area of one bar = | 113.097 | mm2 |
| Number of bars required, $N_{bar}$ = | $\frac{A_s}{A_{bar}}$ = | 14 | |

**Because the number of bars is rounded up, make sure new reinforcement ratio < $\rho_{max}$**

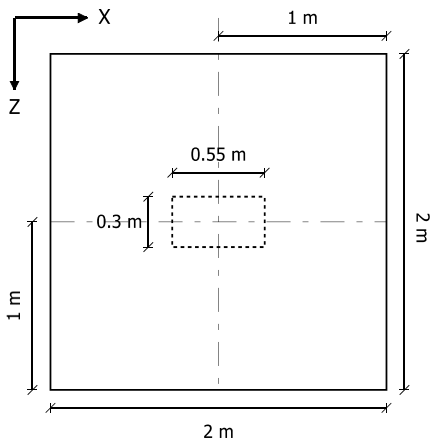
| | | | |
|---|---|---|---|
| Total reinforcement area, $A_{s_total}$ = | $N_{bar}$ X (Area of one bar) = | 1583.352 | mm2 |
| $d_{eff}$ = | D - $C_{cover}$ - 1.5 X (dia. of one bar) = | 0.332 | m |
| Reinforcement ratio, ρ = | $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = | 0.00238 | |

From ACI Cl.7.6.1, minimum req'd clear distance between bars

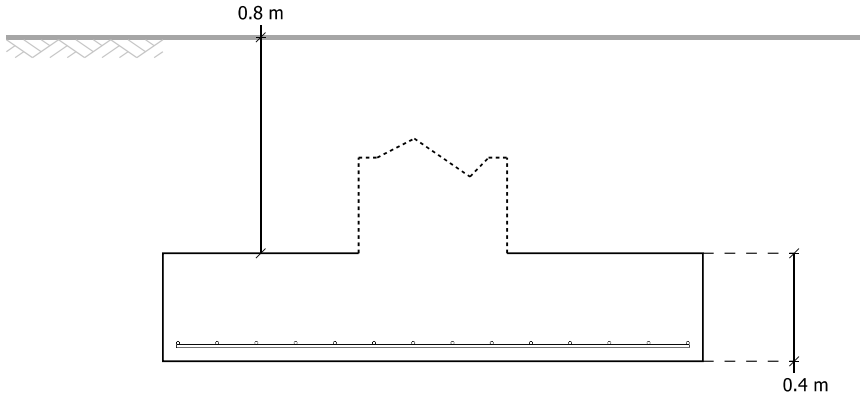
$C_d$ = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.190

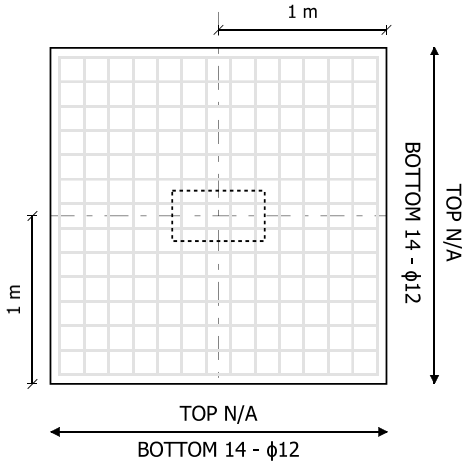
**Check to see if width is sufficient to accomodate bars**




Plan




Elevation




Plan

| | Dimensions (m) | | | Reinforcement | | | |
|---|---|---|---|---|---|---|---|
| | L | B | D | AstX(T) | AstZ(T) | AstX(B) | AstZ(B) |
| Foundation | 2 m | 2 m | 0.4 m | N/A | N/A | 14 - ϕ12 | 14 - ϕ12 |
| | l | b | h | Main Reinforcement | | Trans Reinforcement | |
| Pedestal | N/A | N/A | N/A | N/A | | N/A | |

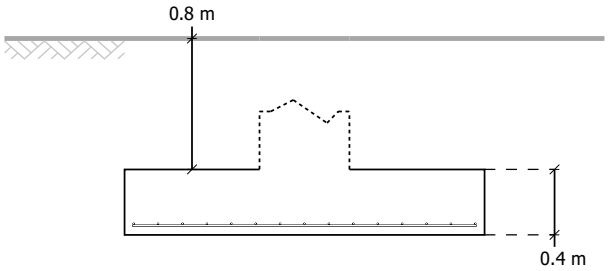
Print Calculation Sheet

# Isolated Footing Design(ACI 318-08)

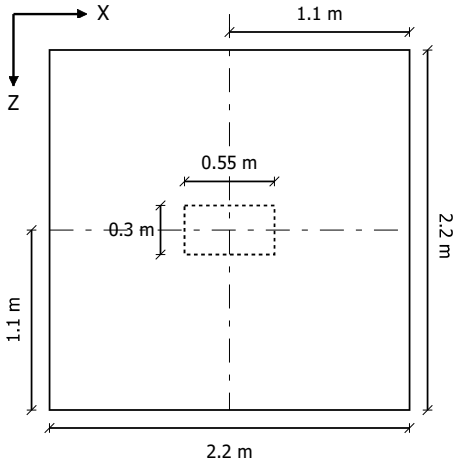
| Footing No. | Group ID | Foundation Geometry | | |
|---|---|---|---|---|
| - | - | Length | Width | Thickness |
| 78 | 1 | 2.200m | 2.200m | 0.400m |

| Footing No. | Footing Reinforcement | | | | Pedestal Reinforcement | |
|---|---|---|---|---|---|---|
| - | Bottom Reinforcement($M_z$) | Bottom Reinforcement($M_x$) | Top Reinforcement($M_z$) | Top Reinforcement($M_x$) | Main Steel | Trans Steel |
| 78 | 15 - ϕ12 | 15 - ϕ12 | N/A | N/A | N/A | N/A |

FOUNDATION-F2




Elevation




Plan

## Input Values

### Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2200.000mm
Footing Width - Z (Fw) : 2200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

### Column Dimensions

Column Shape : Rectangular
Column Length - X ($D_{col}$) : 0.550m

Column Width - Z ($B_{col}$) : 0.300m

## Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

# Design Parameters

## Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

## Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 500.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

## Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

## Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

---

# Design Calculations

## Footing Size

Initial Length ($L_o$) = 2.200m

Initial Width ($W_o$) = 2.200m

## Load Combinations

| Load Combination/s- Service Stress Level | | | | |
|---|---|---|---|---|
| Load Combination Number | Load Combination Title | Load Combination Factor | Soil Bearing Factor | Self Weight Factor |
| 9 | SLS: (DL+T) | 1.00 | 1.00 | 1.00 |
| 10 | SLS: (D+LR) | 1.00 | 1.00 | 1.00 |
| 11 | SLS: (D+0.75LR) | 1.00 | 1.00 | 1.00 |
| 12 | SLS: (D+0.7EZ) | 1.00 | 1.00 | 1.00 |
| 13 | SLS: (D+0.7EX) | 1.00 | 1.00 | 1.00 |
| 14 | SLS: (D+0.75WX+0.75LR) | 1.00 | 1.00 | 1.00 |
| 15 | SLS: (D+0.75WZ+0.75LR) | 1.00 | 1.00 | 1.00 |
| 16 | SLS: (0.6D+WX) | 1.00 | 1.00 | 1.00 |
| 17 | SLS: (0.6D+WZ) | 1.00 | 1.00 | 1.00 |
| 18 | SLS: (0.6D+0.7EX) | 1.00 | 1.00 | 1.00 |
| 19 | SLS: (0.6D+0.7EZ) | 1.00 | 1.00 | 1.00 |
| 20 | SLS: (D+WX) | 1.00 | 1.00 | 1.00 |
| 21 | SLS: (D+WZ) | 1.00 | 1.00 | 1.00 |
| 35 | SLS: (DL+W-Z) | 1.00 | 1.00 | 1.00 |
| 36 | SLS: 0.6DL+WZ- | 1.00 | 1.00 | 1.00 |
| 77 | WL-Z | 1.00 | 1.00 | 1.00 |

| Load Combination/s- Strength Level | | | | |
|---|---|---|---|---|
| Load Combination Number | Load Combination Title | Load Combination Factor | Soil Bearing Factor | Self Weight Factor |
| 22 | ULS: (1.4 DL) | 1.00 | 1.00 | 1.00 |
| 23 | ULS: (1.4D+1.4T+0.5LR) | 1.00 | 1.00 | 1.00 |
| 24 | ULS: (1.4D+1.4T+0.5LR) | 1.00 | 1.00 | 1.00 |
| 25 | ULS: (1.2D+0.80WX+1.6LR) | 1.00 | 1.00 | 1.00 |
| 26 | ULS: (1.2D+1.6WZ+0.5LR) | 1.00 | 1.00 | 1.00 |
| 27 | ULS: (1.2D+1.6WX+0.5LR) | 1.00 | 1.00 | 1.00 |
| 28 | ULS: (1.2D+1.6WZ+0.5LR) | 1.00 | 1.00 | 1.00 |
| 29 | ULS: (1.2D+EX) | 1.00 | 1.00 | 1.00 |
| 30 | ULS: (1.2D+EZ) | 1.00 | 1.00 | 1.00 |
| 31 | ULS: (0.90D+1.6WX) | 1.00 | 1.00 | 1.00 |
| 32 | ULS: (0.90D+1.6WZ) | 1.00 | 1.00 | 1.00 |
| 33 | ULS: (0.90D+EZ) | 1.00 | 1.00 | 1.00 |
| 34 | ULS: (0.90D+EX) | 1.00 | 1.00 | 1.00 |
| 37 | ULS: (0.9DL+1.6WZ-VE) | 1.00 | 1.00 | 1.00 |
| 77 | WL-Z | 1.00 | 1.00 | 1.00 |

## Applied Loads

| Applied Loads - Service Stress Level | | | | | |
|---|---|---|---|---|---|
| LC | Axial (kN) | Shear X (kN) | Shear Z (kN) | Moment X (kNm) | Moment Z (kNm) |
| 9 | 210.079 | -7.777 | -0.152 | 3.115 | -3.993 |
| 10 | 217.164 | -5.295 | -0.412 | 2.506 | 3.368 |
| 11 | 215.181 | -5.497 | -0.403 | 2.444 | 3.457 |
| 12 | 209.237 | -6.087 | 3.779 | 17.161 | 3.695 |
| 13 | 210.649 | -0.269 | -0.441 | 2.072 | -6.174 |
| 14 | 215.626 | -2.462 | 16.205 | 42.987 | -0.692 |
| 15 | 215.175 | -3.742 | 13.780 | 37.538 | 1.953 |
| 16 | 126.134 | 0.385 | 21.919 | 55.413 | -3.299 |
| 17 | 125.532 | -1.322 | 18.685 | 48.147 | 0.228 |
| 18 | 126.956 | 2.172 | -0.291 | 1.168 | -7.662 |
| 19 | 125.544 | -3.646 | 3.929 | 16.257 | 2.206 |
| 20 | 209.827 | -2.056 | 21.769 | 56.317 | -1.810 |
| 21 | 209.225 | -3.763 | 18.535 | 49.051 | 1.717 |
| 35 | 209.203 | -1.467 | -24.225 | -56.576 | -0.219 |
| 36 | 125.510 | 0.975 | -24.075 | -57.480 | -1.708 |
| 77 | -0.030 | 4.636 | -23.850 | -58.836 | -3.941 |

| Applied Loads - Strength Level | | | | | |
|---|---|---|---|---|---|
| LC | Axial (kN) | Shear X (kN) | Shear Z (kN) | Moment X (kNm) | Moment Z (kNm) |
| 22 | 292.926 | -8.544 | -0.525 | 3.164 | 5.211 |
| 23 | 298.076 | -10.483 | -0.231 | 4.484 | -5.767 |
| 24 | 298.076 | -10.483 | -0.231 | 4.484 | -5.767 |
| 25 | 264.243 | -2.793 | 17.206 | 46.351 | -0.525 |
| 26 | 255.032 | -3.176 | 29.788 | 77.701 | 1.082 |
| 27 | 255.995 | -0.444 | 34.962 | 89.326 | -4.561 |
| 28 | 255.032 | -3.176 | 29.788 | 77.701 | 1.082 |
| 29 | 253.102 | 1.011 | -0.544 | 2.443 | -9.670 |
| 30 | 251.085 | -7.301 | 5.484 | 23.999 | 4.428 |
| 31 | 189.260 | 0.982 | 35.093 | 88.525 | -5.501 |
| 32 | 188.297 | -1.749 | 29.919 | 76.900 | 0.142 |
| 33 | 188.315 | -5.470 | 5.597 | 23.321 | 3.311 |
| 34 | 190.332 | 2.841 | -0.432 | 1.765 | -10.787 |
| 37 | 188.262 | 1.925 | -38.498 | -92.104 | -2.956 |
| 77 | -0.030 | 4.636 | -23.850 | -58.836 | -3.941 |

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.840m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.968m^2$

**Note: $A_{min}$ is an initial estimation.**
**P = Critical Factored Axial Load(without self weight/buoyancy/soil).**
**$q_{max}$ = Respective Factored Bearing Capacity.**

## Final Footing Size

Length ($L_2$) = 2.200 m Governing Load Case : # 9

Width ($W_2$) = 2.200 m Governing Load Case : # 9

Depth ($D_2$) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area ($A_2$) = 4.840 $m^2$

Final Soil Height = 0.800 m

Footing Self Weight = 46.462 kN

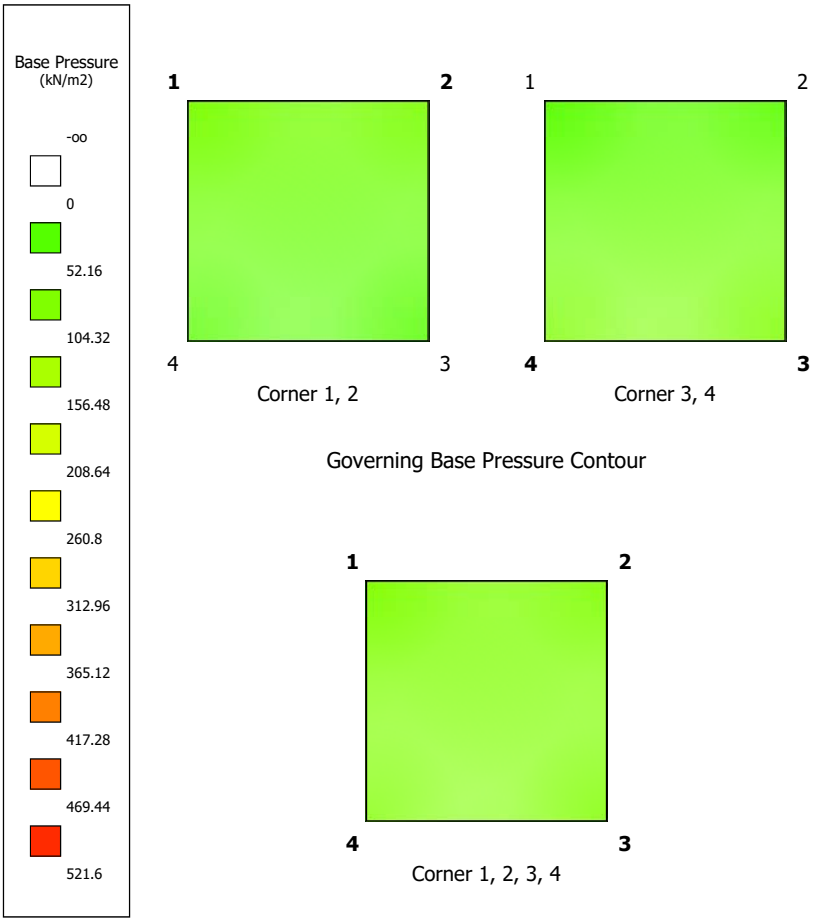
Gross Soil Bearing Capacity = 521.60 kN/m2

Soil Weight On Top Of Footing = 67.318 kN

## Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



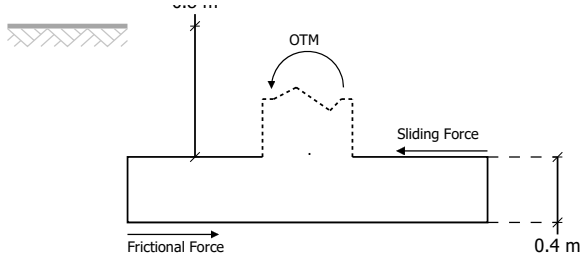

| Load Case | Pressure at corner 1 ($q_1$) (kN/m2) | Pressure at corner 2 ($q_2$) (kN/m2) | Pressure at corner 3 ($q_3$) (kN/m2) | Pressure at corner 4 ($q_4$) (kN/m2) | Area of footing in uplift ($A_u$) ($m^2$) |
|---|---|---|---|---|---|
| 35 | **104.2799** | 103.8659 | 29.1842 | 29.5982 | 0.000 |
| 35 | 104.2799 | **103.8659** | 29.1842 | 29.5982 | 0.000 |
| 20 | 29.6638 | 30.7765 | **104.0580** | 102.9453 | 0.000 |
| 20 | 29.6638 | 30.7765 | 104.0580 | **102.9453** | 0.000 |

If $A_u$ is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

| Load Case | Pressure at corner 1 ($q_1$) (kN/m2) | Pressure at corner 2 ($q_2$) (kN/m2) | Pressure at corner 3 ($q_3$) (kN/m2) | Pressure at corner 4 ($q_4$) (kN/m2) |
|---|---|---|---|---|
| 35 | **104.2799** | 103.8659 | 29.1842 | 29.5982 |
| 35 | 104.2799 | **103.8659** | 29.1842 | 29.5982 |
| 20 | 29.6638 | 30.7765 | **104.0580** | 102.9453 |
| 20 | 29.6638 | 30.7765 | 104.0580 | **102.9453** |

Stability Check




| - | Factor of safety against sliding | | | Factor of safety against overturning | | | |
|---|---|---|---|---|---|---|---|
| Load Case No. | Along X-Direction | Along Z-Direction | Resultant | Required FOS | About X-Direction | About Z-Direction | Required FOS |
| 9 | 22.072 | 1131.972 | 31.209 | 1.500 | 117.265 | 403.921 | 1.500 |
| 10 | 33.127 | 425.863 | 46.707 | 1.500 | 156.295 | 66.392 | 1.500 |
| 11 | 31.718 | 432.989 | 44.736 | 1.500 | 159.300 | 64.020 | 1.500 |
| 12 | 28.123 | 45.304 | 33.791 | 1.500 | 19.130 | 58.001 | 1.500 |
| 13 | 638.998 | 389.926 | 470.720 | 1.500 | 189.311 | 58.867 | 1.500 |
| 14 | 70.922 | 10.773 | 15.063 | 1.500 | 7.363 | 1238.641 | 1.500 |
| 15 | 46.592 | 12.652 | 17.267 | 1.500 | 8.449 | 104.947 | 1.500 |
| 16 | 330.134 | 5.801 | 8.203 | 1.500 | 4.141 | 76.498 | 1.500 |
| 17 | 95.948 | 6.788 | 9.576 | 1.500 | 4.767 | 347.998 | 1.500 |
| 18 | 58.740 | 438.520 | 82.336 | 1.500 | 253.720 | 31.065 | 1.500 |
| 19 | 34.787 | 32.284 | 33.466 | 1.500 | 14.872 | 71.897 | 1.500 |
| 20 | 83.419 | 7.879 | 11.093 | 1.500 | 5.504 | 360.741 | 1.500 |
| 21 | 45.492 | 9.236 | 12.800 | 1.500 | 6.326 | 110.330 | 1.500 |
| 35 | 116.714 | 7.066 | 9.975 | 1.500 | 5.390 | 967.805 | 1.500 |
| 36 | 130.140 | 5.268 | 7.444 | 1.500 | 3.950 | 125.568 | 1.500 |
| 77 | 13.003 | 2.528 | 3.509 | 1.500 | 1.858 | 21.626 | 1.500 |

## Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 77
Governing Disturbing Force : 4.636kN
Governing Restoring Force : 60.287kN
Minimum Sliding Ratio for the Critical Load Case : 13.003
Critical Load Case for Overturning about X-Direction : 77
Governing Overturning Moment : -68.376kNm
Governing Resisting Moment : 127.023kNm
Minimum Overturning Ratio for the Critical Load Case : 1.858

## Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 77
Governing Disturbing Force : -23.850kN
Governing Restoring Force : 60.287kN
Minimum Sliding Ratio for the Critical Load Case : 2.528
Critical Load Case for Overturning about Z-Direction : 77
Governing Overturning Moment : -5.796kNm
Governing Resisting Moment : 125.340kNm

Minimum Overturning Ratio for the Critical Load Case : 21.626

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 77

Governing Disturbing Force : 24.297kN

Governing Restoring Force : 85.259kN

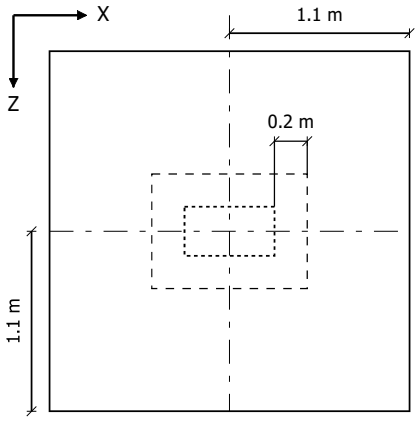
Minimum Sliding Ratio for the Critical Load Case : 3.509

## Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

## Shear Calculation

### Punching Shear Check




Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, $d_{eff}$ = D - $C_{cover}$ - 0.5 * $d_b$ = 0.344m

For rectangular column, $\beta_c$ = $B_{col}$ / $D_{col}$ = 1.833

Effective depth, $d_{eff}$, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 362.865kN, Load Case # 23

From ACI Cl.11.11.2, $b_o$ for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.076m

Equation 11-31, $V_{c1}$ = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2173.720kN

Equation 11-32, $V_{c2}$ = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3364.859kN

Equation 11-33, $V_{c3}$ = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2079.210kN

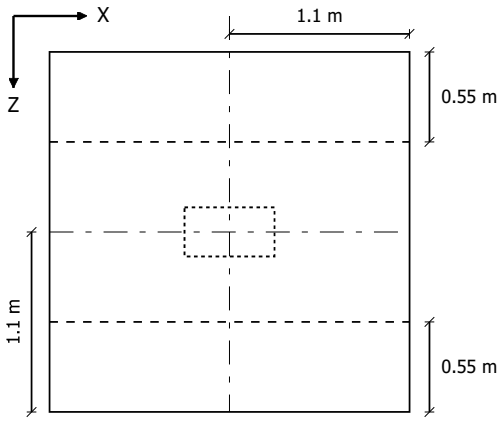
Punching shear strength, $V_c$ = 0.75 X minimum of ($V_{c1}$, $V_{c2}$, $V_{c3}$) = 1559.408kN

0.75 X $V_c$ > $V_u$ hence, OK

### One-Way Shear Check

#### One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)




Plan

From ACI Cl.11.2.1.1, $V_c$ = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 743.541kN

Distance along X to design for shear, $D_x$ = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.606m

Check that 0.75 X $V_c$ > $V_{ux}$ where $V_{ux}$ is the shear force for the critical load cases at a distance $d_{eff}$ from the face of the column caused by bending about the X axis.

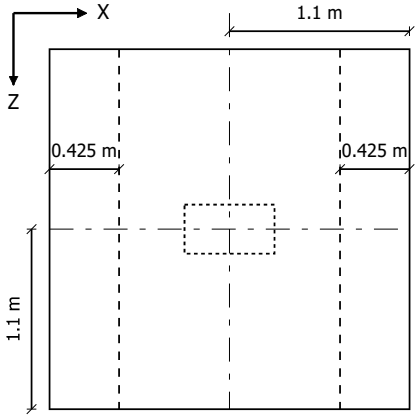
From above calculations, 0.75 X $V_c$ = 557.655 kN

Critical load case for $V_{ux}$ is # 27 $V_{ux} = V_{ux}|_{x=D_x}$ = 158.090 kN

0.75 X $V_c$ > $V_{ux}$ hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)




Plan

From ACI Cl.11.2.1.1, $V_c$ =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 743.541 kN

Distance along X to design for shear, $D_z$ = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.481 m

Check that **0.75 X $V_c$ > $V_{uz}$** where $V_{uz}$ is the shear force for the critical load cases at a distance $d_{eff}$ from the face of the column caused by bending about the Z axis.

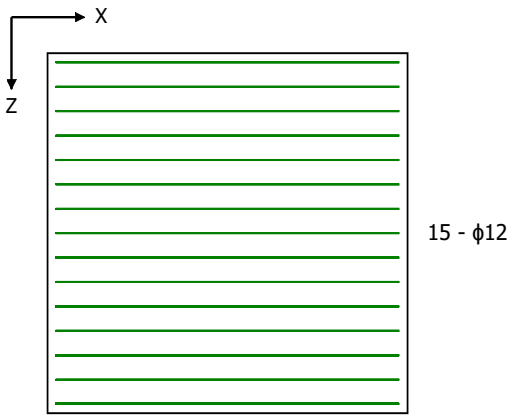
From above calculations, 0.75 X $V_c$ = 557.655 kN

Critical load case for $V_{uz}$ is # 22 $V_{uz} = V_{uz}|_{z=D_z}$ = 92.941 kN

0.75 X $V_c$ > $V_{uz}$ hence, OK

Flexure About Z-Axis

## Design For Bottom Reinforcement Parallel to X Axis




Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 22

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

| | | | |
|---|---|---|---|
| Effective Depth $d_{eff}$ = | | 0.344 | m |
| Factor $\beta_1$ from ACI Cl.10.2.7.3 = | | 0.796 | |
| From ACI Cl. 10.3.2, $\rho_{bal}$ = | $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ | 0.03317 | |
| From ACI Cl. 10.3.3, $\rho_{max}$ = | $0.75 \times \rho_{bal}$ = | 0.02488 | |
| From ACI Cl. 7.12.2, $\rho_{min}$ = | | 0.00155 | |
| From Ref. 1, Eq. 3.8.4a, constant m = | $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = | 14.118 | |

**Calculate reinforcement ratio ρ for critical load case**

| | | | |
|---|---|---|---|
| Design for flexure about Z axis is performed at the face of the column at a distance, $D_x$ = | $0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ = | 0.825 | m |
| Ultimate moment, | $M_u\big\vert_{z=D_z}$ = | 65.641 | kNm |
| Nominal moment capacity, $M_n$ = | $\frac{M_u}{\Phi}$ = | 72.934 | kNm |
| (Based on effective depth) Required ρ = | $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = | 0.00067 | |
| | (Based on gross depth) ρ x $d_{eff}$ / Depth = | 0.00058 | |
| Since | ρ= $\rho_{min}$ | $\rho_{min}$ Governs | |
| Area of Steel Required, $A_s$ = | $\rho \times W \times d_{eff}$ = | 1362.928 | mm2 |

Selected bar Size = #12

Minimum spacing allowed ($S_{min}$) = = 100.000mm

Selected spacing (S) = 150.000mm

$S_{min}$<= S <= $S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

**Based on spacing reinforcement increment; provided reinforcement is**

**#12 @ 150mm o.c.**

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, $D_L$ = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.775 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar}$ = $\frac{A_s}{A_{bar}}$ = 15

**Because the number of bars is rounded up, make sure new reinforcement ratio < $\rho_{max}$**

Total reinforcement area, $A_{s_total}$ = $N_{bar}$ X (Area of one bar) = 1696.448 mm2

$d_{eff}$ = D - $C_{cover}$ - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho$ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00224

From ACI Cl.7.6.1, minimum req'd clear distance between bars

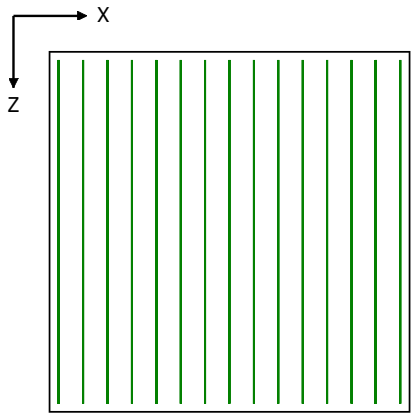
$C_d$ = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.245

**Check to see if width is sufficient to accomodate bars**

## Flexure About X-Axis

### Design For Bottom Reinforcement Parallel to Z Axis




15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth $d_{eff}$ = 0.332 m

Factor $\beta_1$ from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal}$ = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max}$ = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, $\rho_{min}$ = 0.00162

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

**Calculate reinforcement ratio ρ for critical load case**

| | | | |
|---|---|---|---|
| Design for flexure about X axis is performed at the face of the column at a distance, $D_z$ = | $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = | 1.250 | m |
| Ultimate moment, | $M_u\big\vert_{x=D_x}$ = | 117.030 | kNm |
| Nominal moment capacity, $M_n$ = | $\frac{M_u}{\Phi}$ = | 130.033 | kNm |
| (Based on effective depth) Required ρ = | $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^{\ 2}\right)}}\right]$ = | 0.00129 | |
| | (Based on gross depth) ρ x $d_{eff}$ / Depth = | 0.00107 | |
| Since | ρ= $\rho_{min}$ | $\rho_{min}$ Governs | |
| Area of Steel Required, $A_s$ = | $\rho \times W \times d_{eff}$ = | 1428.597 | mm2 |

Selected Bar Size = #12

Minimum spacing allowed ($S_{min}$) = 100.000mm

Selected spacing (S) = 150.000mm

$S_{min}$<= S <= $S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

**Based on spacing reinforcement increment; provided reinforcement is**

**#12 @ 150mm o.c.**

| | | | |
|---|---|---|---|
| Required development length for bars = | $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = | =0.410 | m |
| Available development length for bars, $D_L$ = | $0.5 \times \left(L - D_{col}\right) \cdot - C_{cover}$ = | 0.900 | m |
| Try bar size # 12 | Area of one bar = | 113.097 | mm2 |
| Number of bars required, $N_{bar}$ = | $\frac{A_s}{A_{bar}}$ = | 15 | |

**Because the number of bars is rounded up, make sure new reinforcement ratio < $\rho_{max}$**

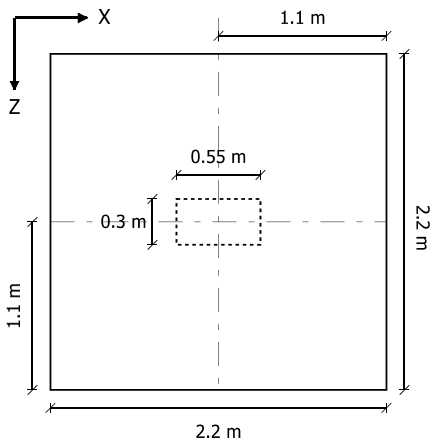
| | | | |
|---|---|---|---|
| Total reinforcement area, $A_{s_total}$ = | $N_{bar}$ X (Area of one bar) = | 1696.448 | mm2 |
| $d_{eff}$ = | D - $C_{cover}$ - 1.5 X (dia. of one bar) = | 0.332 | m |
| Reinforcement ratio, ρ = | $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = | 0.00232 | |

From ACI Cl.7.6.1, minimum req'd clear distance between bars

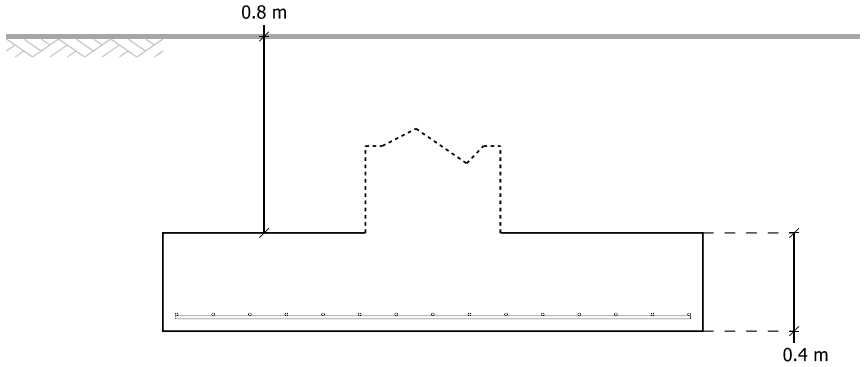
$C_d$ = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.187

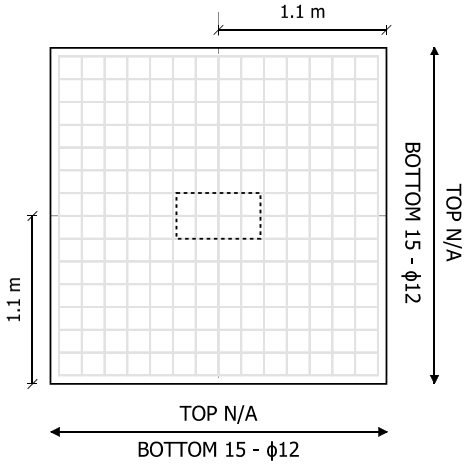
**Check to see if width is sufficient to accomodate bars**




Plan




Elevation




Plan

| | Dimensions (m) | | | Reinforcement | | | |
|---|---|---|---|---|---|---|---|
| Foundation | L | B | D | AstX(T) | AstZ(T) | AstX(B) | AstZ(B) |
| | 2.2 m | 2.2 m | 0.4 m | N/A | N/A | 15 - ϕ12 | 15 - ϕ12 |
| Pedestal | l | b | h | Main Reinforcement | | Trans Reinforcement | |
| | N/A | N/A | N/A | N/A | | N/A | |

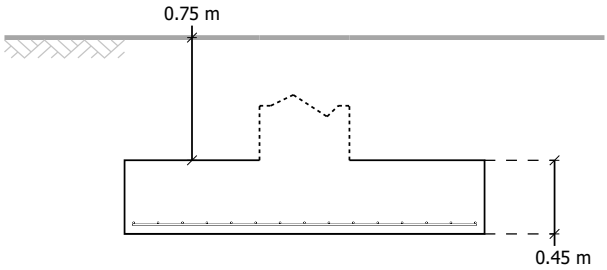
Print Calculation Sheet

# Isolated Footing Design(ACI 318-08)

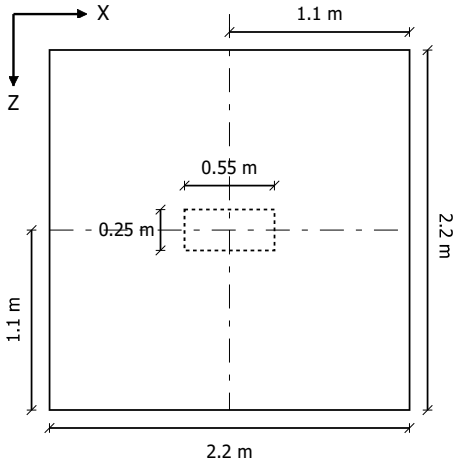
| Footing No. | Group ID | Foundation Geometry | | |
|---|---|---|---|---|
| - | - | Length | Width | Thickness |
| 99 | 1 | 2.200m | 2.200m | 0.450m |

| Footing No. | Footing Reinforcement | | | | Pedestal Reinforcement | |
|---|---|---|---|---|---|---|
| - | Bottom Reinforcement($M_z$) | Bottom Reinforcement($M_x$) | Top Reinforcement($M_z$) | Top Reinforcement($M_x$) | Main Steel | Trans Steel |
| 99 | 15 - ϕ12 | 15 - ϕ12 | N/A | N/A | N/A | N/A |

FOUNDATION-F3




Elevation




Plan

## Input Values

### Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 2200.000mm
Footing Width - Z (Fw) : 2200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

### Column Dimensions

Column Shape : Rectangular
Column Length - X ($D_{col}$) : 0.550m

Column Width - Z ($B_{col}$) : 0.250m

## Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

# Design Parameters

## Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

## Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 500.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

## Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

## Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

---

# Design Calculations

## Footing Size

Initial Length ($L_o$) = 2.200m

Initial Width ($W_o$) = 2.200m

## Load Combinations

| Load Combination/s- Service Stress Level | | | | |
|---|---|---|---|---|
| Load Combination Number | Load Combination Title | Load Combination Factor | Soil Bearing Factor | Self Weight Factor |
| 9 | SLS: (DL+T) | 1.00 | 1.00 | 1.00 |
| 10 | SLS: (D+LR) | 1.00 | 1.00 | 1.00 |
| 11 | SLS: (D+0.75LR) | 1.00 | 1.00 | 1.00 |
| 12 | SLS: (D+0.7EZ) | 1.00 | 1.00 | 1.00 |
| 13 | SLS: (D+0.7EX) | 1.00 | 1.00 | 1.00 |
| 14 | SLS: (D+0.75WX+0.75LR) | 1.00 | 1.00 | 1.00 |
| 15 | SLS: (D+0.75WZ+0.75LR) | 1.00 | 1.00 | 1.00 |
| 16 | SLS: (0.6D+WX) | 1.00 | 1.00 | 1.00 |
| 17 | SLS: (0.6D+WZ) | 1.00 | 1.00 | 1.00 |
| 18 | SLS: (0.6D+0.7EX) | 1.00 | 1.00 | 1.00 |
| 19 | SLS: (0.6D+0.7EZ) | 1.00 | 1.00 | 1.00 |
| 20 | SLS: (D+WX) | 1.00 | 1.00 | 1.00 |
| 21 | SLS: (D+WZ) | 1.00 | 1.00 | 1.00 |
| 35 | SLS: (DL+W-Z) | 1.00 | 1.00 | 1.00 |
| 36 | SLS: 0.6DL+WZ- | 1.00 | 1.00 | 1.00 |
| 77 | WL-Z | 1.00 | 1.00 | 1.00 |

| Load Combination/s- Strength Level | | | | |
|---|---|---|---|---|
| Load Combination Number | Load Combination Title | Load Combination Factor | Soil Bearing Factor | Self Weight Factor |
| 22 | ULS: (1.4 DL) | 1.00 | 1.00 | 1.00 |
| 23 | ULS: (1.4D+1.4T+0.5LR) | 1.00 | 1.00 | 1.00 |
| 24 | ULS: (1.4D+1.4T+0.5LR) | 1.00 | 1.00 | 1.00 |
| 25 | ULS: (1.2D+0.80WX+1.6LR) | 1.00 | 1.00 | 1.00 |
| 26 | ULS: (1.2D+1.6WZ+0.5LR) | 1.00 | 1.00 | 1.00 |
| 27 | ULS: (1.2D+1.6WX+0.5LR) | 1.00 | 1.00 | 1.00 |
| 28 | ULS: (1.2D+1.6WZ+0.5LR) | 1.00 | 1.00 | 1.00 |
| 29 | ULS: (1.2D+EX) | 1.00 | 1.00 | 1.00 |
| 30 | ULS: (1.2D+EZ) | 1.00 | 1.00 | 1.00 |
| 31 | ULS: (0.90D+1.6WX) | 1.00 | 1.00 | 1.00 |
| 32 | ULS: (0.90D+1.6WZ) | 1.00 | 1.00 | 1.00 |
| 33 | ULS: (0.90D+EZ) | 1.00 | 1.00 | 1.00 |
| 34 | ULS: (0.90D+EX) | 1.00 | 1.00 | 1.00 |
| 37 | ULS: (0.9DL+1.6WZ-VE) | 1.00 | 1.00 | 1.00 |
| 77 | WL-Z | 1.00 | 1.00 | 1.00 |

## Applied Loads

| Applied Loads - Service Stress Level | | | | | |
|---|---|---|---|---|---|
| LC | Axial (kN) | Shear X (kN) | Shear Z (kN) | Moment X (kNm) | Moment Z (kNm) |
| 9 | 202.928 | 3.408 | 0.000 | -0.008 | -4.391 |
| 10 | 216.959 | 3.666 | 0.000 | 0.002 | -4.735 |
| 11 | 213.455 | 3.603 | 0.000 | 0.002 | -4.653 |
| 12 | 202.940 | 3.412 | 6.898 | 30.251 | -4.407 |
| 13 | 202.513 | 10.347 | 0.000 | 0.006 | -23.067 |
| 14 | 213.305 | 6.226 | 0.000 | 0.009 | -11.713 |
| 15 | 213.429 | 3.637 | -0.002 | -0.003 | -4.748 |
| 16 | 121.565 | 5.546 | 0.000 | 0.010 | -12.057 |
| 17 | 121.731 | 2.093 | -0.002 | -0.005 | -2.771 |
| 18 | 121.337 | 8.982 | 0.000 | 0.005 | -21.304 |
| 19 | 121.764 | 2.047 | 6.898 | 30.250 | -2.644 |
| 20 | 202.741 | 6.911 | 0.000 | 0.011 | -13.820 |
| 21 | 202.907 | 3.458 | -0.002 | -0.004 | -4.534 |
| 35 | 202.878 | 3.513 | 0.003 | 0.008 | -4.685 |
| 36 | 121.702 | 2.148 | 0.003 | 0.007 | -2.922 |
| 77 | -0.063 | 0.101 | 0.003 | 0.006 | -0.277 |

| Applied Loads - Strength Level | | | | | |
|---|---|---|---|---|---|
| LC | Axial (kN) | Shear X (kN) | Shear Z (kN) | Moment X (kNm) | Moment Z (kNm) |
| 22 | 284.118 | 4.777 | 0.000 | 0.003 | -6.171 |
| 23 | 291.108 | 4.899 | 0.001 | -0.011 | -6.312 |
| 24 | 291.108 | 4.899 | 0.001 | -0.011 | -6.312 |
| 25 | 265.798 | 7.300 | 0.000 | 0.010 | -13.343 |
| 26 | 250.484 | 4.295 | -0.004 | -0.008 | -5.655 |
| 27 | 250.218 | 9.819 | 0.000 | 0.017 | -20.513 |
| 28 | 250.484 | 4.295 | -0.004 | -0.008 | -5.655 |
| 29 | 242.918 | 14.001 | 0.000 | 0.008 | -31.946 |
| 30 | 243.528 | 4.094 | 9.854 | 43.216 | -5.289 |
| 31 | 182.327 | 8.668 | 0.000 | 0.016 | -19.027 |
| 32 | 182.593 | 3.144 | -0.004 | -0.008 | -4.169 |
| 33 | 182.646 | 3.071 | 9.854 | 43.215 | -3.966 |
| 34 | 182.036 | 12.978 | 0.000 | 0.008 | -30.623 |
| 37 | 182.546 | 3.233 | 0.004 | 0.011 | -4.410 |
| 77 | -0.063 | 0.101 | 0.003 | 0.006 | -0.277 |

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.840m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.816m^2$

**Note: $A_{min}$ is an initial estimation.**
**P = Critical Factored Axial Load(without self weight/buoyancy/soil).**
**$q_{max}$ = Respective Factored Bearing Capacity.**

## Final Footing Size

Length ($L_2$) = 2.200 m Governing Load Case : # 9

Width ($W_2$) = 2.200 m Governing Load Case : # 9

Depth ($D_2$) = 0.450 m Governing Load Case : # 30

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area ($A_2$) = 4.840 $m^2$

Final Soil Height = 0.750 m

Footing Self Weight = 52.270 kN

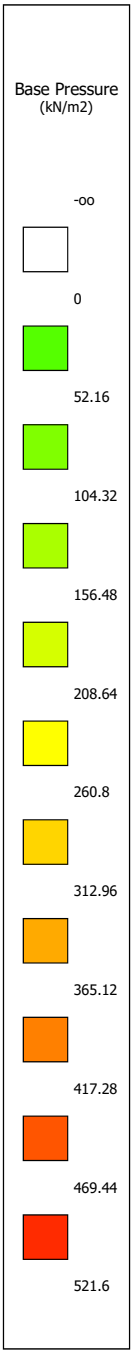
Gross Soil Bearing Capacity = 521.60 kN/m2

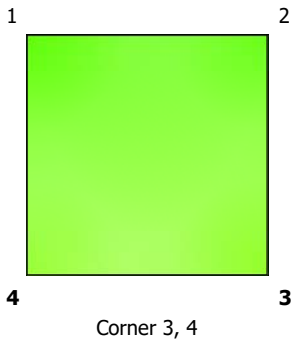
Soil Weight On Top Of Footing = 63.482 kN

## Pressures at 4 Corners

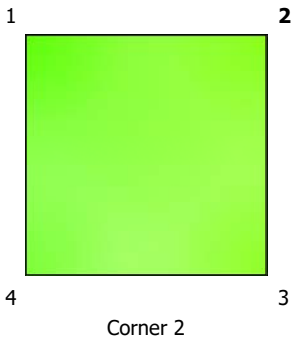
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

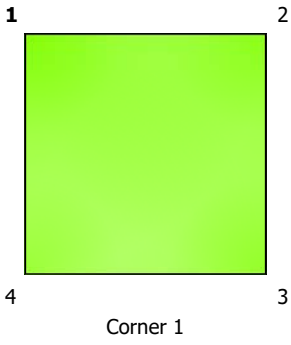



Corner 3, 4

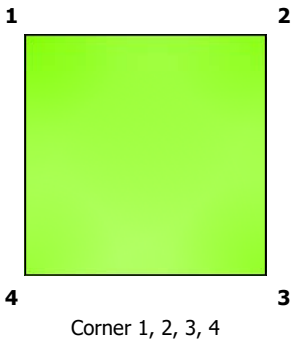



Corner 2




Corner 1

Governing Base Pressure Contour




Corner 1, 2, 3, 4

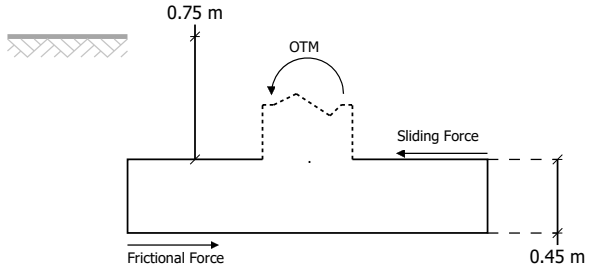
| Load Case | Pressure at corner 1 ($q_1$) (kN/m2) | Pressure at corner 2 ($q_2$) (kN/m2) | Pressure at corner 3 ($q_3$) (kN/m2) | Pressure at corner 4 ($q_4$) (kN/m2) | Area of footing in uplift ($A_u$) ($m^2$) |
|---|---|---|---|---|---|
| 10 | **65.1427** | 72.3385 | 72.3411 | 65.1453 | 0.000 |
| 13 | 50.1318 | **81.3756** | 81.3826 | 50.1388 | 0.000 |
| 12 | 43.7013 | 50.3986 | **87.9895** | 81.2923 | 0.000 |
| 12 | 43.7013 | 50.3986 | 87.9895 | **81.2923** | 0.000 |

If $A_u$ is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

| Load Case | Pressure at corner 1 ($q_1$) (kN/m2) | Pressure at corner 2 ($q_2$) (kN/m2) | Pressure at corner 3 ($q_3$) (kN/m2) | Pressure at corner 4 ($q_4$) (kN/m2) |
|---|---|---|---|---|
| 10 | **65.1427** | 72.3385 | 72.3411 | 65.1453 |
| 13 | 50.1318 | **81.3756** | 81.3826 | 50.1388 |
| 12 | 43.7013 | 50.3986 | **87.9895** | 81.2923 |
| 12 | 43.7013 | 50.3986 | 87.9895 | **81.2923** |

## Stability Check




| - | Factor of safety against sliding | | | Factor of safety against overturning | | | |
|---|---|---|---|---|---|---|---|
| Load Case No. | Along X-Direction | Along Z-Direction | Resultant | Required FOS | About X-Direction | About Z-Direction | Required FOS |
| 9 | 49.555 | N/A | 70.081 | 1.500 | 44307.506 | 59.127 | 1.500 |
| 10 | 48.100 | N/A | 68.024 | 1.500 | 164005.717 | 57.286 | 1.500 |
| 11 | 48.432 | N/A | 68.493 | 1.500 | 164264.649 | 57.680 | 1.500 |
| 12 | 49.504 | 24.487 | 31.040 | 1.500 | 10.557 | 58.956 | 1.500 |
| 13 | 16.302 | N/A | 23.055 | 1.500 | 57061.247 | 12.621 | 1.500 |
| 14 | 28.010 | N/A | 39.612 | 1.500 | 40823.022 | 24.923 | 1.500 |
| 15 | 47.970 | 102566.597 | 67.840 | 1.500 | 106665.704 | 56.682 | 1.500 |
| 16 | 22.680 | N/A | 32.074 | 1.500 | 25746.274 | 17.924 | 1.500 |
| 17 | 60.133 | 51520.925 | 85.040 | 1.500 | 42255.915 | 70.307 | 1.500 |
| 18 | 13.990 | N/A | 19.784 | 1.500 | 49399.891 | 10.281 | 1.500 |
| 19 | 61.494 | 18.250 | 24.743 | 1.500 | 7.880 | 73.221 | 1.500 |
| 20 | 24.426 | N/A | 34.544 | 1.500 | 31842.622 | 20.681 | 1.500 |
| 21 | 48.840 | 71928.937 | 69.070 | 1.500 | 65596.770 | 57.524 | 1.500 |
| 35 | 48.068 | 59990.713 | 67.979 | 1.500 | 37947.250 | 55.904 | 1.500 |
| 36 | 58.580 | 46268.505 | 82.844 | 1.500 | 31185.908 | 67.118 | 1.500 |
| 77 | 606.737 | 23793.186 | 857.777 | 1.500 | 18022.062 | 393.839 | 1.500 |

## Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 8.982kN
Governing Restoring Force : 125.657kN
Minimum Sliding Ratio for the Critical Load Case : 13.990
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 33.354kNm
Governing Resisting Moment : 262.840kNm
Minimum Overturning Ratio for the Critical Load Case : 7.880

## Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19
Governing Disturbing Force : 6.898kN
Governing Restoring Force : 125.883kN
Minimum Sliding Ratio for the Critical Load Case : 18.250
Critical Load Case for Overturning about Z-Direction : 18
Governing Overturning Moment : -25.346kNm
Governing Resisting Moment : 260.588kNm
Minimum Overturning Ratio for the Critical Load Case : 10.281

## Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

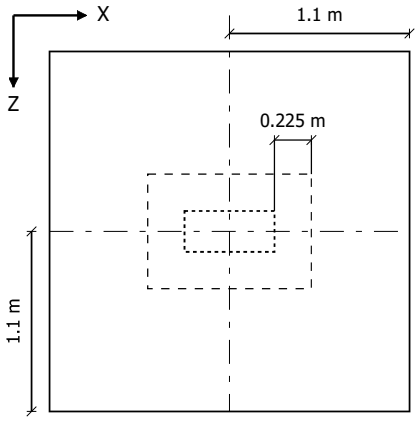
Critical Load Case for Sliding along Resultant Direction : 18
Governing Disturbing Force : 8.982kN
Governing Restoring Force : 177.706kN
Minimum Sliding Ratio for the Critical Load Case : 19.784

## Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

## Shear Calculation

### Punching Shear Check




Plan

Total Footing Depth, D = 0.450m
Calculated Effective Depth, $d_{eff}$ = D - $C_{cover}$ - 0.5 * $d_b$ = 0.394m
For rectangular column, $\beta_c$ = $B_{col}$ / $D_{col}$ = 2.200

Effective depth, $d_{eff}$, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 355.755kN, Load Case # 23

From ACI Cl.11.11.2, $b_o$ for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, $V_{c1}$ = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2347.075kN

Equation 11-32, $V_{c2}$ = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4279.745kN

Equation 11-33, $V_{c3}$ = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2458.840kN

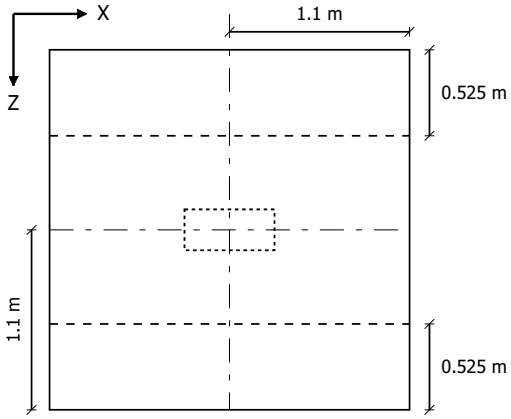
Punching shear strength, $V_c$ = 0.75 X minimum of ($V_{c1}$, $V_{c2}$, $V_{c3}$) = 1760.306kN

0.75 X $V_c$ > $V_u$ hence, OK

### One-Way Shear Check

## One-Way Shear Along X

## (Shear Plane Parallel to Global X Axis)




Plan

From ACI Cl.11.2.1.1, $V_c$ = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 851.613kN

Distance along X to design for shear, $D_x$ = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.581m

Check that 0.75 X $V_c$ > $V_{ux}$ where $V_{ux}$ is the shear force for the critical load cases at a distance $d_{eff}$ from the face of the column caused by bending about the X axis.

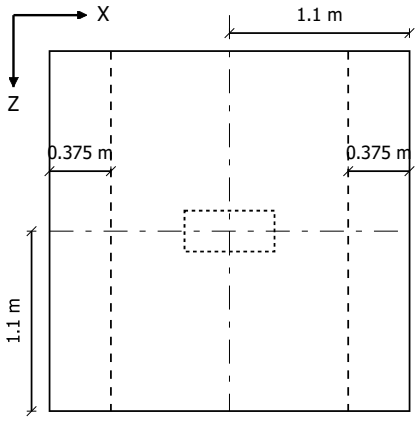
From above calculations, 0.75 X $V_c$ = 638.710 kN

Critical load case for $V_{ux}$ is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 120.139 kN

0.75 X $V_c$ > $V_{ux}$ hence, OK

## One-Way Shear Along Z

## (Shear Plane Parallel to Global Z Axis)




Plan

From ACI Cl.11.2.1.1, $V_c$ =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 851.613 kN

Distance along X to design for shear, $D_z$ = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X $V_c$ > $V_{uz}$** where $V_{uz}$ is the shear force for the critical load cases at a distance $d_{eff}$ from the face of the column caused by bending about the Z axis.

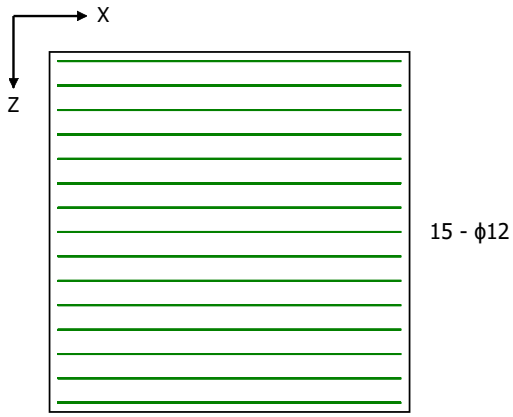
From above calculations, 0.75 X $V_c$ = 638.710 kN

Critical load case for $V_{uz}$ is # 29 $V_{uz} = V_{uz}|_{z=D_z}$ = 86.698 kN

0.75 X $V_c$ > $V_{uz}$ hence, OK

## Flexure About Z-Axis

### Design For Bottom Reinforcement Parallel to X Axis




Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

| | | |
|---|---|---|
| Effective Depth $d_{eff}$ = | | 0.394 m |
| Factor $\beta_1$ from ACI Cl.10.2.7.3 = | | 0.796 |
| From ACI Cl. 10.3.2, $\rho_{bal}$ = | $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ | 0.03317 |
| From ACI Cl. 10.3.3, $\rho_{max}$ = | $0.75 \times \rho_{bal}$ = | 0.02488 |
| From ACI Cl. 7.12.2, $\rho_{min}$ = | | 0.00160 |
| From Ref. 1, Eq. 3.8.4a, constant m = | $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = | 14.118 |

**Calculate reinforcement ratio ρ for critical load case**

| | | |
|---|---|---|
| Design for flexure about Z axis is performed at the face of the column at a distance, $D_x$ = | $0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ = | 1.375 m |
| Ultimate moment, | $M_u\big\vert_{z=D_z}$ = | 67.597 kNm |
| Nominal moment capacity, $M_n$ = | $\frac{M_u}{\Phi}$ = | 75.108 kNm |
| (Based on effective depth) Required ρ = | $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = | 0.00053 |
| | (Based on gross depth) ρ x $d_{eff}$ / Depth = | 0.00046 |
| Since | ρ= $\rho_{min}$ | $\rho_{min}$ Governs |
| Area of Steel Required, $A_s$ = | $\rho \times W \times d_{eff}$ = | 1580.784 mm2 |

Selected bar Size = #12

Minimum spacing allowed ($S_{min}$) = = 100.000mm

Selected spacing (S) = 150.000mm

$S_{min}$<= S <= $S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

**Based on spacing reinforcement increment; provided reinforcement is**

| **#12 @ 150mm o.c.** |
|---|

| | | | |
|---|---|---|---|
| Required development length for bars = | $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c'}}$ = | =0.410 | m |
| Available development length for bars, $D_L$ = | $0.5 \times (L - D_{col}) - C_{cover}$ = | 0.775 | m |
| Try bar size # 12 | Area of one bar = | 113.097 | mm2 |
| Number of bars required, $N_{bar}$ = | $\frac{A_s}{A_{bar}}$ = | 15 | |

**Because the number of bars is rounded up, make sure new reinforcement ratio < $\rho_{max}$**

| | | | |
|---|---|---|---|
| Total reinforcement area, $A_{s_total}$ = | $N_{bar}$ X (Area of one bar) = | 1696.448 | mm2 |
| $d_{eff}$ = | D - $C_{ccover}$ - 0.5 X (dia. of one bar) = | 0.394 | m |
| Reinforcement ratio, $\rho$ = | $\frac{A_{s_total}}{(d_{eff} \times W)}$ = | 0.00196 | |

From ACI Cl.7.6.1, minimum req'd clear distance between bars

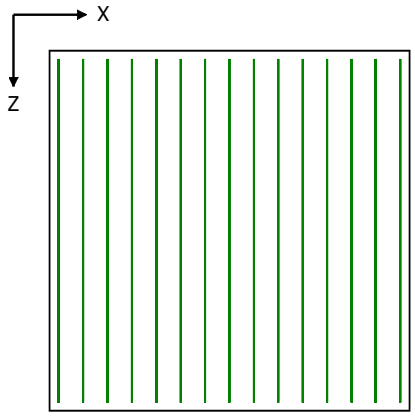
$C_d$ = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.073

**Check to see if width is sufficient to accomodate bars**

## Flexure About X-Axis

### Design For Bottom Reinforcement Parallel to Z Axis




15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 30

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

| | | | |
|---|---|---|---|
| Effective Depth $d_{eff}$ = | | 0.382 | m |
| Factor $\beta_1$ from ACI Cl.10.2.7.3 = | | 0.796 | |
| From ACI Cl. 10.3.2, $\rho_{bal}$ = | $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ | 0.03317 | |
| From ACI Cl. 10.3.3, $\rho_{max}$ = | $0.75 \times \rho_{bal}$ = | 0.02488 | |

From ACI Cl.7.12.2, $\rho_{min}$ = 0.00166

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

**Calculate reinforcement ratio $\rho$ for critical load case**

Design for flexure about X axis is performed at the face of the column at a distance, $D_z$ = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.225 m

Ultimate moment, $M_u|_{x=D_x}$ = 97.416 kNm

Nominal moment capacity, $M_n$ = $\frac{M_u}{\Phi}$ = 108.240 kNm

(Based on effective depth) Required $\rho$ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00081

(Based on gross depth) $\rho$ x $d_{eff}$ / Depth = 0.00069

Since $\rho = \rho_{min}$ $\rho_{min}$ Governs

Area of Steel Required, $A_s$ = $\rho \times W \times d_{eff}$ = 1646.513 mm2

Selected Bar Size = #12

Minimum spacing allowed ($S_{min}$) = 100.000mm

Selected spacing (S) = 149.143mm

$S_{min}$<= S <= $S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 250.272mm

Safe for Cracking Aspect.

**Based on spacing reinforcement increment; provided reinforcement is**

| **#12 @ 145mm o.c.** |
|---|

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, $D_L$ = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.925 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar}$ = $\frac{A_s}{A_{bar}}$ = 15

**Because the number of bars is rounded up, make sure new reinforcement ratio < $\rho_{max}$**

Total reinforcement area, $A_{s_total}$ = $N_{bar}$ X (Area of one bar) = 1696.448 mm2

$d_{eff}$ = D - $C_{cover}$ - 1.5 X (dia. of one bar) = 0.382 m

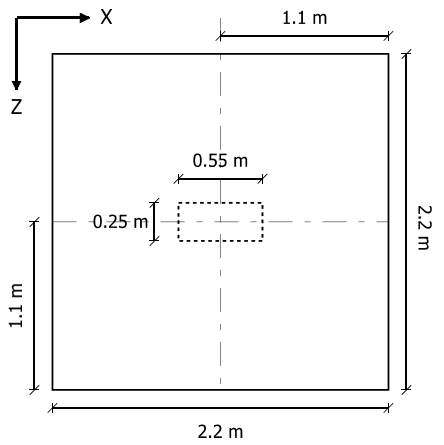
Reinforcement ratio, $\rho$ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00202

From ACI Cl.7.6.1, minimum req'd clear distance between bars

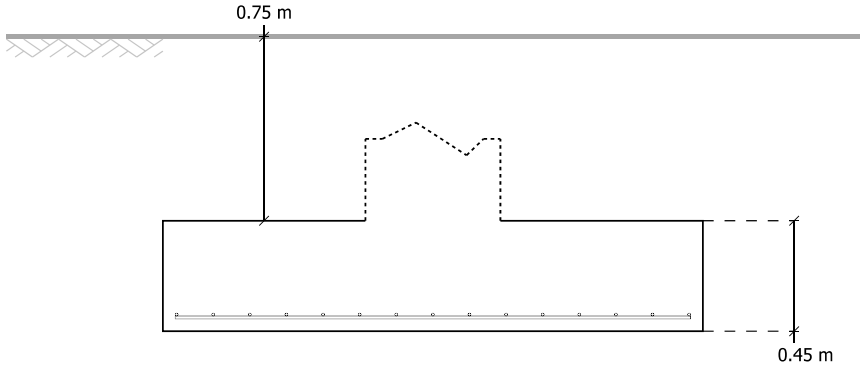
$C_d$ = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.030

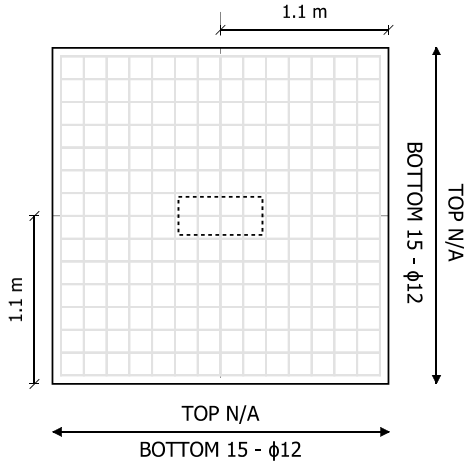
**Check to see if width is sufficient to accomodate bars**




Plan




Elevation




Plan

| | Dimensions (m) | | | Reinforcement | | | |
|---|---|---|---|---|---|---|---|
| | L | B | D | AstX(T) | AstZ(T) | AstX(B) | AstZ(B) |
| Foundation | 2.2 m | 2.2 m | 0.45 m | N/A | N/A | 15 - ϕ12 | 15 - ϕ12 |
| | l | b | h | Main Reinforcement | | Trans Reinforcement | |
| Pedestal | N/A | N/A | N/A | N/A | | N/A | |

Print Calculation Sheet